CFA Level I Mock Exam 5.2 - Mark Scheme

Each question lists the correct choice and the original feedback for all three choices. Source excerpts accompany explanations where PDF text extraction may lose equation layout.

Question 1

Two years ago, a homeowner took out a $1 million home mortgage from a bank. The current principal on the loan is $750,000, and the homeowner has defaulted on the loan. Following foreclosure proceedings, the bank sells the property for $600,000 and is only entitled to use these funds to satisfy the loan obligation. The homeowner most likely had a:

Correct answer: C

A. bullet loan. - Incorrect

Incorrect Answer Feedback: Incorrect because a bullet loan is a type of interest-only mortgage loan in which there are no principal payments over the term of the loan. The loan balance outstanding is less than the original mortgage amount, so it is unlikely that this was a bullet loan.

B. recourse loan. - Incorrect

Incorrect Answer Feedback: Incorrect because in the case of a recourse loan the bank would have been entitled to make a claim for the shortfall of $150,000 against the borrower.

C. non-recourse loan. - Correct

Correct Answer Feedback: Correct because the bank does not have a claim against the borrower for the shortfall of $150,000 on the mortgage balance outstanding relative to the proceeds received from the property's sale, indicating that the home mortgage is a non-recourse loan.

Question 2

Valuing private equity investments using estimated prices rather than market prices most likely causes an:

Correct answer: B

A. overestimation of both return volatility and the correlation of returns with traditional asset classes. - Incorrect

Incorrect Answer Feedback: Incorrect because these measures will be understated not overstated. Measuring historical private equity performance is challenging; as with hedge funds, which will be discussed later, private equity return indexes typically rely on self- reporting and are subject to survivorship, backfill, and other biases. This typically leads to an overstatement of returns. Moreover, prior to the global financial crisis of 2008–2009, in the absence of a liquidity event, private equity firms did not necessarily mark their investments to market. Failure to mark to market combined with the lag to mark investments due to inherent illiquidity will understate measures of volatility and correlations with other investments.

B. underestimation of both return volatility and the correlation of returns with traditional asset classes. - Correct

Correct Answer Feedback: Correct because measuring historical private equity performance is challenging; as with hedge funds, which will be discussed later, private equity return indexes typically rely on self-reporting and are subject to survivorship, backfill, and other biases. This typically leads to an overstatement of returns. Moreover, prior to the global financial crisis of 2008–2009, in the absence of a liquidity event, private equity firms did not necessarily mark their investments to market. Failure to mark to market combined with the lag to mark investments due to inherent illiquidity will understate measures of volatility and correlations with other investments.

C. underestimation of return volatility and overestimation of the correlation of returns with traditional asset classes. - Incorrect

Incorrect Answer Feedback: Incorrect because both of these measures will be understated. Measuring historical private equity performance is challenging; as with hedge funds, which will be discussed later, private equity return indexes typically rely on self-reporting and are subject to survivorship, backfill, and other biases. This typically

leads to an overstatement of returns. Moreover, prior to the global financial crisis of 2008–2009, in the absence of a liquidity event, private equity firms did not necessarily mark their investments to market. Failure to mark to market combined with the lag to mark investments due to inherent illiquidity will understate measures of volatility and correlations with other investments.

Question 3

A greenfield infrastructure investment is best described as an asset that has:

Correct answer: A

A. yet to be constructed. - Correct

Correct Answer Feedback: Correct because greenfield investments entail developing new assets and new infrastructure.

B. already been constructed. - Incorrect

Incorrect Answer Feedback: Incorrect because greenfield investments entail developing new assets and new infrastructure. An investment on an asset that has already been constructed is most likely considered a brownfield investment as those investments expand existing facilities and may involve privatization of public assets or a sale leaseback of completed greenfield projects.

C. reached the end of its useful life. - Incorrect

Incorrect Answer Feedback: Incorrect because greenfield investments entail developing new assets and new infrastructure. An investment on an asset that has already been constructed is most likely considered a brownfield investment as those investments expand existing facilities and may involve privatization of public assets or a sale leaseback of completed greenfield projects.

Question 4

Which of the following statements is most accurate? US commercial paper typically:

Correct answer: B

A. requires the issuer to pledge collateral. - Incorrect

Incorrect Answer Feedback: Incorrect because commercial paper is a short-term, unsecured promissory note issued in the public market or via a private placement that represents a debt obligation of the issuer. Being unsecured, the issuer is not required to pledge collateral.

B. requires the issuer to have a backup line of credit. - Correct

Correct Answer Feedback: Correct because credit rating agencies often require that commercial paper issuers secure a backup line of credit from banks.

C. has a maturity ranging from a few days up to two years. - Incorrect

Incorrect Answer Feedback: Incorrect because maturities for US commercial paper typically range from a few days up to 270 days.

Question 5

A limit order is an example of a(n):

Correct answer: C

A. validity instruction. - Incorrect

Incorrect Answer Feedback: Incorrect because validity instructions indicate when the order may be filled. B.

B. clearing instruction. - Incorrect

Incorrect Answer Feedback: Incorrect because clearing instructions indicate how to arrange the final settlement of the trade.

C. execution instruction. - Correct

Correct Answer Feedback: Correct because execution instructions indicate how to fill the order. Market and limit orders convey the most common execution instructions.

Question 6

Direct ownership of commercial real estate most likely:

Correct answer: C

A. provides greater liquidity than a real estate investment trust. - Incorrect

Incorrect Answer Feedback: Incorrect because it is typically difficult to quickly buy or sell direct investments in real estate, and transaction costs are typically high. REITs and other private real estate funds are structured as infinite-life, open-end funds and allow investors to contribute or redeem capital throughout the life of the fund in a fashion parallel to mutual fund structures.

B. requires a lower initial investment than a real estate investment trust. - Incorrect

Incorrect Answer Feedback: Incorrect because investing directly in property requires significant capital. The owners need to have access to a potentially significant amount of debt and equity capital because of the large initial capital outlay needed for real estate investments. REITS are a form of indirect real estate investments which require lower capital outlay.

C. offers greater investor control over underlying properties than a real estate investment trust. - Correct

Correct Answer Feedback: Correct because only the owner can decide when to buy or sell, when and how much to spend on capital projects, whom to select as tenants based on credit quality preference and tenant mix, and what types of lease terms to offer. Owners generate cash flow returns from the use and enjoyment of the property, the receipt of lease payments, and the potential for capital appreciation. REITS are a form of indirect real estate investments, in which investors have less control over underlying properties.

Question 7

A net investment hedge occurs when a derivative is used to:

Correct answer: C

A. offset the fluctuation in the fair value of an asset or liability. - Incorrect

Incorrect Answer Feedback: Incorrect because this describes a fair value hedge, not a net investment hedge. A fair value hedge designation applies when a derivative is deemed to offset the fluctuation in fair value of an asset or liability.

B. absorb the variable cash flow of a floating rate asset or liability. - Incorrect

Incorrect Answer Feedback: Incorrect because this describes a cash flow hedge, not a net investment hedge. Derivatives designated as absorbing the variable cash flow of a floating-rate asset or liability such as foreign exchange, interest rates, or commodities are referred to as cash flow hedges.

C. offset the exchange rate risk of the equity of a foreign operation. - Correct

Correct Answer Feedback: Correct because net investment hedges occur when either a foreign currency bond or a derivative such as an FX swap or forward is used to offset the exchange rate risk of the equity of a foreign operation.

Question 8

Consider bonds that have the same yield-to-maturity and maturity. The bond with the greatest reinvestment risk is most likely the one selling at:

Correct answer: C

A. a discount. - Incorrect

Incorrect Answer Feedback: Incorrect because the bond selling at a discount has a lower coupon rate than the bond selling at a premium.

B. par. - Incorrect

Incorrect Answer Feedback: Incorrect because the bond selling at par has a lower coupon rate than the bond selling at a premium.

C. a premium. - Correct

Correct Answer Feedback: Correct because yield-to-maturity is based on the assumption that a bond is held to maturity, does not default, and has its coupon payments reinvested at the yield-to-maturity. The bond selling at a premium has the highest coupon rate and is expected to earn the most reinvestment income from reinvesting those coupon payments at the yield- to-maturity. If the reinvestment rate falls, this bond will suffer the greatest loss.

Question 9

Which of the following is most likely considered an investor constraint in an investment policy statement?

Correct answer: A

A. Time horizon - Correct

Correct Answer Feedback: Correct because five major types of constraints on portfolio selection are liquidity, time horizon, tax concerns, legal and regulatory factors, and unique circumstances.

B. Risk tolerance - Incorrect

Incorrect Answer Feedback: Incorrect because client's objectives are specified in terms of risk tolerance and return requirements. Risk and return objectives are, therefore, likely to be part of the investment objectives section of the investment policy statement and not part of the constraints section focusing on liquidity, time horizon, tax concerns, legal and regulatory factors, and unique circumstances.

C. Return requirement - Incorrect

Incorrect Answer Feedback: Incorrect because client's objectives are specified in terms of risk tolerance and return requirements. Risk and return objectives are, therefore, likely to be part of the investment objectives section of the investment policy statement and not part of the constraints section focusing on liquidity, time horizon, tax concerns, legal and regulatory factors, and unique circumstances.

Question 10

A positive duration gap indicates that a bond investor is currently:

Correct answer: B

A. at risk of lower interest rates. - Incorrect

Incorrect Answer Feedback: Incorrect because a negative (not positive) duration gap indicates that the investor is at risk of lower interest rates.

B. at risk of higher interest rates. - Correct

Correct Answer Feedback: Correct because the duration gap is a bond's Macaulay duration minus the investment horizon. A positive duration gap indicates that the investor currently is at risk of higher interest rates.

C. hedged against interest rate risk. - Incorrect

Incorrect Answer Feedback: Incorrect because a duration gap of zero (not a positive duration gap) indicates that the bond investor currently is hedged against interest rate risk.

Question 11

Both equity and fixed-income indexes can be categorized according to the:

Correct answer: B

A. currency of payments. - Incorrect

Incorrect Answer Feedback: Incorrect because this feature is specific to fixed-income indexes.

B. issuer's economic sector. - Correct

Correct Answer Feedback: Correct because both equity and fixed-income indexes can be constructed according to sector. Similar to equities, fixed-income securities can be categorized according to the issuer's economic sector, the issuer's geographic region, or the economic development of the issuer's geographic region.

C. degree of inflation protection. - Incorrect

Incorrect Answer Feedback: Incorrect because this feature is specific to fixed-income indexes.

Question 12

Real estate is best classified as a:

Correct answer: B

A. commodity. - Incorrect

Incorrect Answer Feedback: Incorrect because commodities include precious metals, energy products, industrial metals, and agricultural products.

B. physical asset. - Correct

Correct Answer Feedback: Correct because real assets are tangible properties such as real estate, airplanes, or machinery. Securities, currencies, and contracts are classified as financial assets whereas commodities and real assets (including real estate) are classified as physical assets.

C. financial asset. - Incorrect

Incorrect Answer Feedback: Incorrect because real assets are tangible properties such as real estate, airplanes, or machinery. Securities, currencies, and contracts are classified as financial assets whereas commodities and real assets (including real estate) are classified as physical assets.

Question 13

An analyst gathers the following information to evaluate the effect of dividends and leverage on future growth:

	Scenario 1	Scenario 2
Dividend payout ratio	60%	40%
Financial leverage	3.0	2.5

If return on assets is the same for each scenario, the sustainable growth rate is:

Correct answer: B

A. higher under Scenario 1. - Incorrect

Incorrect Answer Feedback: Incorrect because scenario 2's sustainable growth rate 1.5 × ROA is higher than Scenario 1's sustainable growth rate 1.2 × ROA.

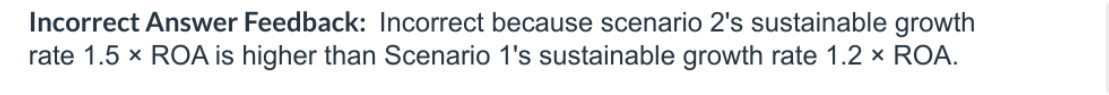


Figure 1: Source equation for question 13, option A

B. higher under Scenario 2. - Correct

Correct Answer Feedback: Correct because sustainable growth rate g = b × ROE, where b = retention rate = (1 – Dividend payout ratio), and ROE = ROA × Financial leverage. Thus g = (1 – Dividend payout ratio) × ROA × Financial leverage. Since ROA is unchanged between Scenario 1 and 2: g = (1 – 0.6) × ROA × 3.0 = 1.2 × ROA; and scenario 1 g = (1 – 0.4) × ROA × 2.5 = 1.5 × ROA. scenario 2 Accordingly, scenario 2's sustainable growth rate 1.5 × ROA is higher than Scenario 1's sustainable growth rate 1.2 × ROA.

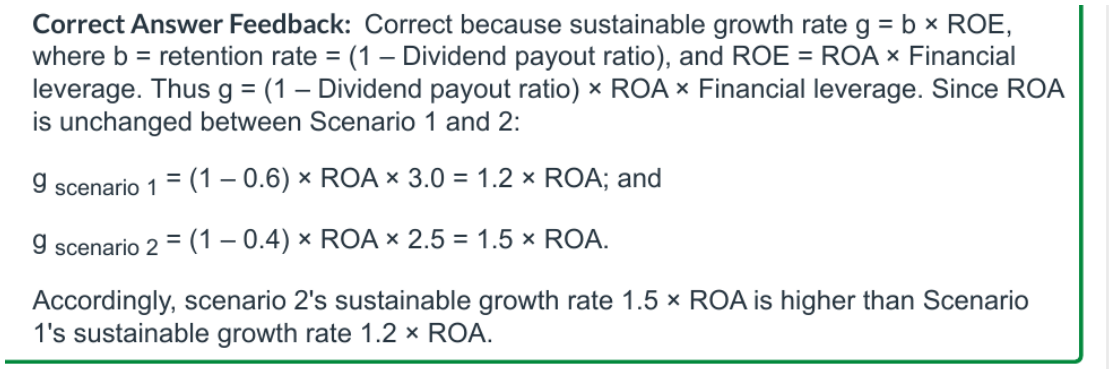


Figure 2: Source equation for question 13, option B

C. the same under both Scenario 1 and Scenario 2. - Incorrect

Incorrect Answer Feedback: Incorrect because scenario 2's sustainable growth rate 1.5 × ROA is higher than Scenario 1's sustainable growth rate 1.2 × ROA.

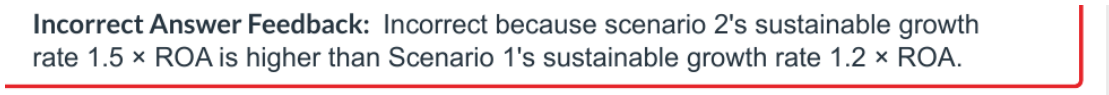


Figure 3: Source equation for question 13, option C

Question 14

An analyst gathers the following data (in INR millions) for a company located in India:

Market value of debt	10.0
Market value of preferred stock	5.0
Cash and short-term investments	4.5
Earnings before interest, taxes, depreciation, and amortization (EBITDA)	15.0

Assuming an enterprise value multiple of 3.2, the company's market capitalization (in INR millions) is closest to:

Correct answer: C

A. 28.5. - Incorrect

Incorrect Answer Feedback: Incorrect because it incorrectly subtracts cash and short- term investments instead of adding them. Enterprise Value (EV) = EBITDA × EV multiple = 15 × 3.2 = 48 Market capitalization = Enterprise value – MV of debt – MV of preferred stock – cash and short-term investments = 48 – 10 – 5 – 4.5 = 28.5

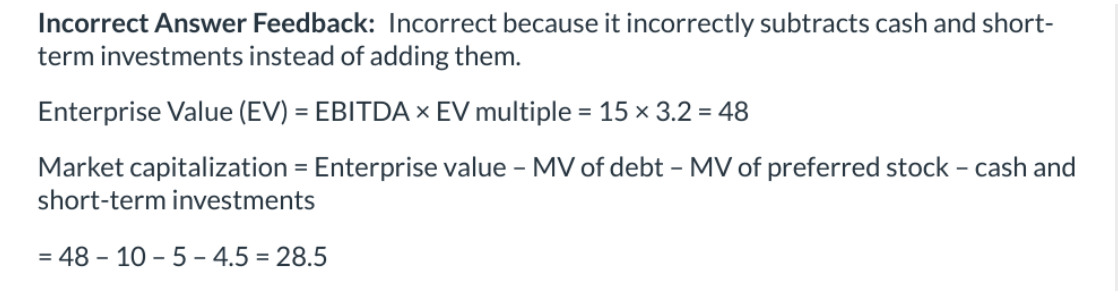


Figure 4: Source equation for question 14, option A

B. 33.0. - Incorrect

Incorrect Answer Feedback: Incorrect. It ignores cash and short-term investments. Enterprise Value (EV) = EBITDA × EV multiple = 15 × 3.2 = 48 Market capitalization = Enterprise value – MV of debt – MV of preferred stock – cash and short-term investments = 48 – 10 – 5 = 33

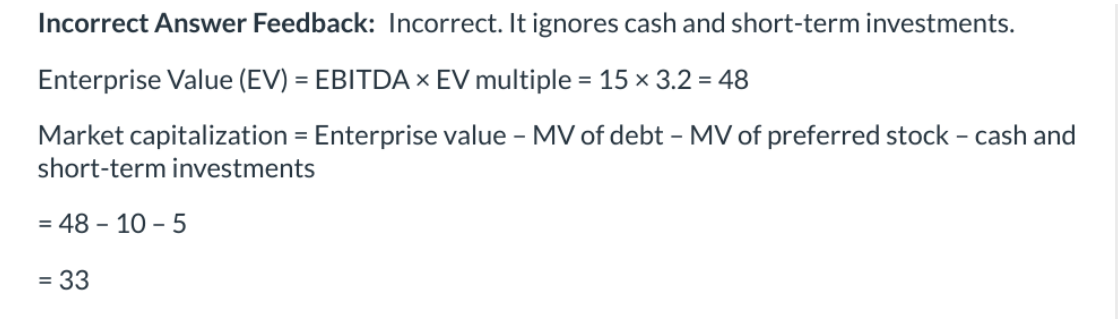


Figure 5: Source equation for question 14, option B

C. 37.5. - Correct

Correct Answer Feedback: Correct because Enterprise value (EV) = EBITDA × EV multiple = 15 × 3.2 = 48. Market capitalization = EV – Market value (MV) of debt – MV of preferred stock + Cash and short-term investments = 48 – 10 – 5 + 4.5 = 37.5.

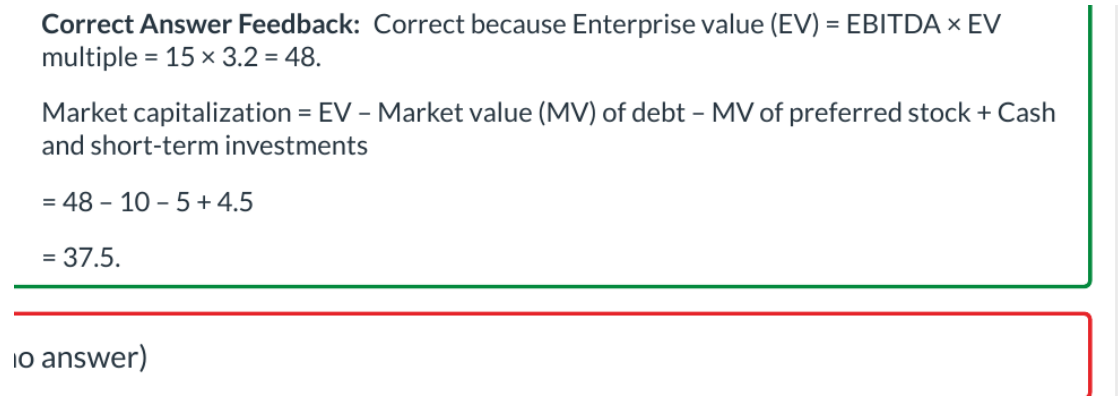


Figure 6: Source equation for question 14, option C

Question 15

Financial forecasting using historical base rates and convergence is best suited for companies that are:

Correct answer: B

A. in highly cyclical industries. - Incorrect

Incorrect Answer Feedback: Incorrect because this approach is less appropriate for companies in changing or new industries for which calculating a base rate is difficult or less relevant, for companies in highly cyclical industries for which a longer-term base rate and smooth convergence to it would obscure year-to-year volatility, and for companies that are industry leaders by a wide margin and account for a substantial proportion of any industry base rate.

B. in well-established industries. - Correct

Correct Answer Feedback: Correct because historical base rates and convergence may be appropriate for forecasting companies in well-established industries with many publicly traded peers, such as banks, airlines, restaurants, automakers, and retailers, especially when the analyst's industry analysis does not call for material changes in industry structure or external industry influences.

C. industry leaders by a wide margin. - Incorrect

Incorrect Answer Feedback: Incorrect because this approach is less appropriate for companies in changing or new industries for which calculating a base rate is difficult or less relevant, for companies in highly cyclical industries for which a longer-term base rate and smooth convergence to it would obscure year-to-year volatility, and for companies that are industry leaders by a wide margin and account for a substantial proportion of any industry base rate.

Question 16

A bond has a modified duration of 7.5 years and an annual convexity of 280. If yields decrease by 60 basis points, the percentage price change is closest to:

Correct answer: B

A. 4.00%. - Incorrect

Incorrect Answer Feedback: Incorrect because the convexity adjustment is subtracted instead of added: ≠ (–7.5 × –0.006) – (0.5 × 280 × 0.0062) = 0.045 – 0.00504 = 0.03996 ≈ 4.00%.

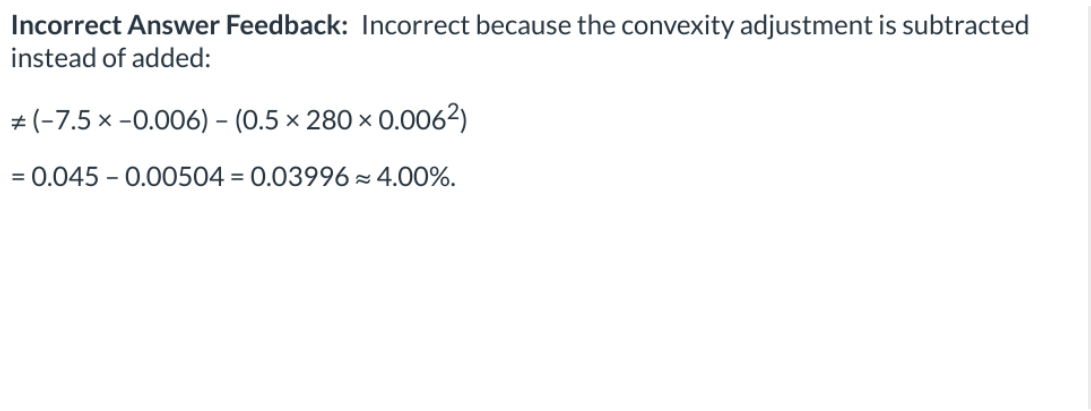


Figure 7: Source equation for question 16, option A

B. 5.00%. - Correct

Correct Answer Feedback: Correct because the %ΔPV = (–AnnModDur × ΔYield) + (0.5 × AnnConvexity × ΔYield2) 2 = (–7.5 × –0.006) + (0.5 × 280 × 0.006) = 0.045 + 0.00504 = 0.05004 ≈ 5.00%.

Correct Answer Feedback: Correct because the %ΔPV = (–AnnModDur × ΔYield) + (0.5 × AnnConvexity × ΔYield$^{2)}$

Figure 8: Source equation for question 16, option B

= 0.045 + 0.00504 = 0.05004 ≈ 5.00%.

Figure 9: Source equation for question 16, option B

C. 5.51%. - Incorrect

Incorrect Answer Feedback: Incorrect because the convexity adjustment is missing the 0.5 in the calculation: ≠ (–7.5 × –0.006) + (280 × 0.0062) = 0.045 + 0.01008 = 0.05508 ≈ 5.51%.

Incorrect Answer Feedback: Incorrect because the convexity adjustment is missing the 0.5 in the calculation:

≠ (–7.5 × –0.006) + (280 × 0.006^2)

= 0.045 + 0.01008 = 0.05508 ≈ 5.51%.

ıo answer)

Figure 10: Source equation for question 16, option C

Question 17

An analyst observes the fee structure and performance of a hedge fund with $100 million assets under management at the beginning of the year:

Annual management fee	2%
Incentive fee	20%
Hard hurdle rate	3%

The annual management fee is based on the year-end valuation, and the incentive fee is calculated net of the management fee. If the fund return before fees is 10% at year end, the total annual fee earned by the fund manager is closest to:

Correct answer: A

A. $3.2 million. - Correct

Correct Answer Feedback: Correct because the hurdle rate is a minimum rate of return, typically 8%, that the GP must exceed in order to earn the performance fee. GPs typically receive 20% of the total profit of the private equity fund net of any hard hurdle rate, in which case the GP earns fees on annual returns in excess of the hurdle rate, or net of the soft hurdle rate, in which case the fee is calculated on the entire annual gross return as long as the set hurdle is exceeded. Management fee = $110 million × 2% = $2.2 million. Annual return is 10%; and the hard hurdle rate is 3%. Thus the incentive fee is paid on a 7% return: 7% × $100 million = $7 million Thus the incentive fee = ($7 million – $.2.2 million management fee) × 20% = $0.96 million Therefore, total fees = management fee + incentive fee = $2.2 million + $0.96 million = $3.16 million, rounded to $3.2 million.

B. $3.6 million. - Incorrect

Incorrect Answer Feedback: Incorrect because the incentive fee is calculated gross of management fee. Annual return is 10%; and the hard hurdle rate is 3%. Thus the incentive fee is paid on a 7% return: 7% × $100 million = $7 million. The incentive fee ≠ $7 million × 20% = $1.4 million. Therefore, total fees ≠ management fee + incentive fee = $2.2 million + $1.4 million = $3.6 million.

C. $3.8 million. - Incorrect

Incorrect Answer Feedback: Incorrect because a soft hurdle rate is used instead of a hard hurdle rate. Manage-

Correct Answer Feedback: Correct because the hurdle rate is a *minimum* rate of return, typically 8%, that the GP must exceed in order to earn the performance fee. GPs typically receive 20% of the total profit of the private equity fund net of any *hard hurdle rate*, in which case the GP earns fees on annual returns in excess of the hurdle rate, or net of the *soft hurdle rate*, in which case the fee is calculated on the entire annual gross return as long as the set hurdle is exceeded.

Management fee = $110 million × 2% = $2.2 million.

Figure 11: Source equation for question 17, option A

Annual return is 10%; and the hard hurdle rate is 3%. Thus the incentive fee is paid on a 7% return: 7% × $100 million = $7 million

Thus the incentive fee = ($7 million – $.2.2 million management fee) × 20% = $0.96 million

Therefore, total fees = management fee + incentive fee = $2.2 million + $0.96 million = $3.16 million, rounded to $3.2 million.

Figure 12: Source equation for question 17, option A

Incorrect Answer Feedback: Incorrect because the incentive fee is calculated gross of management fee.

Annual return is 10%; and the hard hurdle rate is 3%. Thus the incentive fee is paid on a 7% return: 7% × $100 million = $7 million.

The incentive fee

≠

$7 million × 20% = $1.4 million.

Therefore, total fees

≠

management fee + incentive fee = $2.2 million + $1.4 million = $3.6 million.

Figure 13: Source equation for question 17, option B

Incorrect Answer Feedback: Incorrect because a soft hurdle rate is used instead of a hard hurdle rate.

Management fee = $110 million × 2% = $2.2 million.

As the hurdle rate was exceeded, the incentive fee = ($10 million – $2.2 million) × 20% = $1.56 million.

Therefore, total fees = management fee + incentive fee ≠ $2.2 million + $1.56 million = $3.76 million, rounded to $3.8 million.

Figure 14: Source equation for question 17, option C

Question 18

Which of the following measures is least appropriate for evaluating the risk of a currency option?

Correct answer: A

A. Beta - Correct

Correct Answer Feedback: Correct because beta is a measure of the sensitivity of a security's returns to the returns on the market portfolio. Beta measures relative risk, meaning how much market risk an asset contributes to a well-diversified portfolio. Beta describes risk well for a portfolio of equities, but other sources of risk may require other descriptive risk metrics. The risk associated with derivatives is one example of this.

B. Vega - Incorrect

Incorrect Answer Feedback: Incorrect because vega is a first-order measure of the change in the derivative price for a change in the volatility of the underlying. Vega is an appropriate measure of derivatives risk, whereas beta is not.

C. Delta - Incorrect

Incorrect Answer Feedback: Incorrect because the sensitivity of the derivative price to a small change in the value of the underlying asset is called the delta. It is perhaps the most important measure of derivatives risk.

Question 19

Compared to a European waterfall, an American waterfall is:

Correct answer: C

A. less advantageous to the general partner. - Incorrect

Incorrect Answer Feedback: Incorrect because deal-by-deal (or American) waterfalls are more advantageous to the general partner because performance fees are collected on a per-deal basis, allowing the general partner to get paid before limited partners receive their initial investment and their preferred rate of return (i.e., the hurdle rate) on the entire fund.

B. equally advantageous to the general partner. - Incorrect

Incorrect Answer Feedback: Incorrect because deal-by-deal (or American) waterfalls are more advantageous to the general partner because performance fees are collected on a per-deal basis, allowing the general partner to get paid before limited partners receive their initial investment and their preferred rate of return (i.e., the hurdle rate) on the entire fund.

C. more advantageous to the general partner. - Correct

Correct Answer Feedback: Correct because there are two types of waterfalls: deal-by- deal (or American) waterfalls and whole-of-fund (or European) waterfalls. Deal-by-deal waterfalls are more advantageous to the general partner because performance fees are collected on a per-deal basis, allowing the general partner to get paid before limited partners receive their initial investment and their preferred rate of return (i.e., the hurdle rate) on the entire fund. In whole-of-fund waterfalls, all distributions go to the limited partners as deals are exited and the general partner does not participate in any profits until the limited partners receive their initial investment and the hurdle rate has been met. In contrast to deal-by-deal waterfalls, whole-of-fund waterfalls occur at the aggregate fund level (i.e., after all investments have been exited) and are more advantageous to the limited partners.

Question 20

For a given equity index, the long-term total return of the index is most likely greater than the price return:

Correct answer: A

A. for any series of price returns. - Correct

Correct Answer Feedback: Correct because at inception, the values of the price and total return versions of an index are equal. As time passes, however, the value of the total return index, which includes the reinvestment of all dividends and/or interest received, will exceed the value of the price return index by an increasing amount.

B. only when the price returns are less than zero. - Incorrect

Incorrect Answer Feedback: Incorrect because as time passes the value of the total return index, which includes the reinvestment of all dividends and/or interest received, will exceed the value of the price return index by an increasing amount.

C. only when the price returns are equal to or greater than zero. - Incorrect

Incorrect Answer Feedback: Incorrect because as time passes the value of the total return index, which includes the reinvestment of all dividends and/or interest received, will exceed the value of the price return index by an increasing amount.

Question 21

With respect to a client's IPS, which of the following measures is most likely used when stating a relative risk objective?

Correct answer: B

A. Value at risk - Incorrect

Incorrect Answer Feedback: Incorrect because value at risk is used to measure an absolute risk objective, not a relative risk objective. Measures of absolute risk include the variance or standard deviation of returns and value at risk.

B. Tracking risk - Correct

Correct Answer Feedback: Correct because quantitative risk objectives can be absolute or relative or a combination of the two. Some clients may choose to express relative risk objectives, which relate risk relative to one or more benchmarks perceived to represent appropriate risk standards. For risk relative to a benchmark, the measure could be tracking risk, or tracking error.

C. Standard deviation of returns - Incorrect

Incorrect Answer Feedback: Incorrect because standard deviation of returns is used to measure an absolute risk objective, not a relative risk objective. Measures of absolute risk include the variance or standard deviation of returns and value at risk.

Question 22

Hedge funds are most likely characterized by:

Correct answer: C

A. significant investment restrictions. - Incorrect

Incorrect Answer Feedback: Incorrect because hedge funds are also subject to much lighter regulatory, compliance, and transparency requirements. With more flexibility in portfolio construction, hedge funds enjoy leeway to invest in situations in which time may be needed to generate an expected return.

B. a high correlation of returns with investment-grade bonds. - Incorrect

Incorrect Answer Feedback: Incorrect because hedge fund performance has a very low correlation with that of traditional asset classes, such as investment bonds and currencies/cash. (However, hedge funds have a fairly high correlation with equities.)

C. performance relative to a traditional portfolio that may be hard to gauge. - Correct

Correct Answer Feedback: Correct because hedge fund portfolios approach return generation differently from traditional portfolios. The most significant difference is that hedge funds seek to limit market exposure and returns from beta and primarily focus on generating idiosyncratic returns by identifying sources of unique return, or alpha. Due to this different approach to return generation, hedge fund performance relative to a traditional portfolio may be hard to gauge.

Question 23

Insurance on a portfolio of risky assets is most likely:

Correct answer: A

A. negatively correlated with the portfolio's assets. - Correct

Correct Answer Feedback: Correct because insurance has a negative correlation with other assets and is, thus, very valuable. Insurance gives a positive return when other assets lose value, but pays nothing if other assets maintain their value.

B. uncorrelated with the portfolio's assets. - Incorrect

Incorrect Answer Feedback: Incorrect because insurance has a negative correlation with other assets in the risky portfolio.

C. positively correlated with the portfolio's assets. - Incorrect

Incorrect Answer Feedback: Incorrect because insurance has a negative correlation with other assets in the risky portfolio.

Question 24

When forming asset classes for a strategic asset allocation, the defined asset classes should:

Correct answer: A

A. approximate the relevant investable universe. - Correct

Correct Answer Feedback: Correct because asset classes, while being mutually exclusive, should add up to a sufficient approximation of the relevant investable universe.

B. have high pairwise correlations between them. - Incorrect

Incorrect Answer Feedback: Incorrect because an asset class should provide diversification relative to other asset classes. In statistical terms paired correlations of assets should be lower versus assets in other asset classes.

C. contain negatively correlated assets within each asset class. - Incorrect

Incorrect Answer Feedback: Incorrect because paired correlations of assets should be relatively high (not negative) within an asset class.

Question 25

An analyst gathers the following information about a company's common stock:

Dividend per share – Year 1	$2.00
Dividend per share – Year 2	$2.10
Annual dividend per share – Year 3 and beyond	$2.21

If the required rate of return is 10%, the intrinsic value per share is closest to:

Correct answer: A

A. $21.82. - Correct

Correct Answer Feedback: Correct because it is the present value of all future dividends. The present value of the perpetual level dividends, three years from now, is: $2.21 / (0.10 – 0) = $22.10. The present value of the perpetual level dividends now is: $22.10 / (1.10)3 = $16.604. The present of dividends in the first three years is: $2.00 / 2 3 1.10 + $2.10 / (1.10) + $2.21 / (1.10) = $1.818 + $1.736 + $1.660 = $5.214. So, the 2 3 1.10 + $2.10 / (1.10) + $2.21 / (1.10) = $1.818 + $1.736 + $1.660 = $5.214. So, the intrinsic value of the stock is closest to: $16.604 + $5.214 = $21.818 ≈ $21.82.

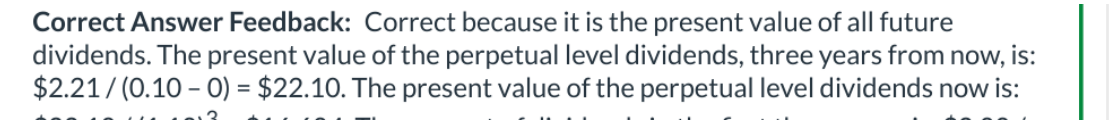


Figure 15: Source equation for question 25, option A

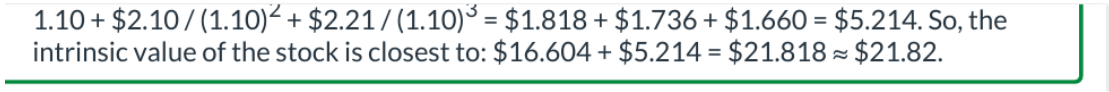


Figure 16: Source equation for question 25, option A

B. $22.10. - Incorrect

Incorrect Answer Feedback: Incorrect because it is only the present value of the perpetual level dividends three years from now. $2.21 / (0.10 – 0) = $22.10.

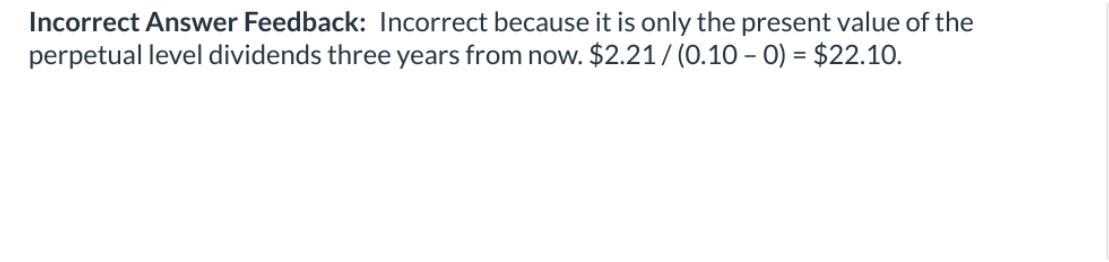


Figure 17: Source equation for question 25, option B

C. $27.31. - Incorrect

Incorrect Answer Feedback: Incorrect because the present value of the level dividends in perpetuity three years from now has been added to the present value of the dividends in Year 1, Year 2 and Year 3, without discounting it for three years (see calculations in correct answer). $22.100 + $5.214 = $27.314 ≈ $27.31.

Question 26

A call option with an exercise price of $38 was bought for $3. The price of the underlying increases from $42 to $47. At expiration, the payoff to the call buyer is equal to:

Correct answer: C

A. $4. - Incorrect

Incorrect Answer Feedback: Incorrect because the correct formula c = Max(0,S – T T X) (payoff to the call buyer) is changed to deduct 38 = X the exercise price from the current price of the underlying S = $42 (instead of the price at 0 expiration $47) to yield $4.

B. $6. - Incorrect

Incorrect Answer Feedback: Incorrect because this is the profit to the call buyer calculated Π = Max(0,S – X) – c (profit to the call buyer) and inputting the values T 0 S = $47 as the price of the underlying at expiration less $38 = X as the T exercise price, and then subtracting the call premium of $3 yields $6.

Incorrect Answer Feedback: Incorrect because the present value of the level dividends in perpetuity three years from now has been added to the present value of the dividends in Year 1, Year 2 and Year 3, without discounting it for three years (see calculations in correct answer). $22.100 + $5.214 = $27.314 ≈ $27.31.

Figure 18: Source equation for question 25, option C

Incorrect Answer Feedback: Incorrect because the correct formula $c_T = Max(0, S_T - X)$ (payoff to the call buyer) is changed to deduct 38 = X the exercise price from the current price of the underlying S_0 = $42 (instead of the price at expiration $47) to yield $4.

Figure 19: Source equation for question 26, option A

C. $9. - Correct

Correct Answer Feedback: Correct because c = Max(0,S – X) (payoff to the T T call buyer) and inputting the values S = $47 as the price of the underlying T at expiration less $38 = X as the exercise price yields $9.

Question 27

According to capital market theory, the only type of risk that is priced is:

Correct answer: A

A. systematic risk. - Correct

Correct Answer Feedback: Correct because systematic or non-diversifiable risk is priced and investors are compensated for holding assets or portfolios based only on that investment's systematic risk. Investors do not receive any return for accepting nonsystematic or diversifiable risk. Pricing or valuing an asset is equivalent to estimating its expected rate of return.

B. diversifiable risk. - Incorrect

Incorrect Answer Feedback: Incorrect because systematic or non-diversifiable risk is priced and investors are compensated for holding assets or portfolios based only on that investment's systematic risk. Investors do not receive any return for accepting nonsystematic or diversifiable risk.

C. idiosyncratic risk. - Incorrect

Incorrect Answer Feedback: Incorrect because systematic or non-diversifiable risk is priced and investors are compensated for holding assets or portfolios based only on that investment's systematic risk. Investors do not receive any return for accepting nonsystematic or diversifiable risk. Nonsystematic risk is the risk that pertains to a single company or industry and is also known as company-specific, industry-specific, diversifiable, or idiosyncratic risk.

Incorrect Answer Feedback: Incorrect because this is the profit to the call buyer calculated $\Pi = Max(0, S_T - X) - c_0$ (profit to the call buyer) and inputting the values S_T = $47 as the price of the underlying at expiration less $38 = X as the exercise price, and then subtracting the call premium of $3 yields $6.

Figure 20: Source equation for question 26, option B

Correct Answer Feedback: Correct because $c_T = Max(0, S_T - X)$ (payoff to the call buyer) and inputting the values S_T = \$47 as the price of the underlying at expiration less \$38 = X as the exercise price yields \$9.

ıo answer)

Figure 21: Source equation for question 26, option C

Question 28

An increase in which of the following most likely increases the time value of a European put option?

Correct answer: A

A. The volatility of the underlying only - Correct

Correct Answer Feedback: Correct because the time value reflects the value of the uncertainty that arises from the volatility in the underlying. Also, time value results in an option price being greater with volatility and time but declining as expiration approaches. Therefore, the increase in the volatility of the underlying will increase the time value of a put option. In contrast, the value of a European put is inversely related to the risk-free interest rate.

B. The current market interest rate only - Incorrect

Incorrect Answer Feedback: Incorrect because the value of a European put is inversely (not directly) related to the risk-free interest rate.

C. Both the volatility of the underlying and the current market interest rate - Incorrect

Incorrect Answer Feedback: Incorrect because the value of a European put is inversely (not directly) related to the risk-free interest rate.

Question 29

An investor gathers the following information:

Risk-free rate per annum	1.50%
Exercise price	\$9.75
9-month call option price	\$3.25

According to put–call–forward parity, the price of a 9-month fiduciary call is closest to:

Correct answer: A

A. \$12.89. - Correct

Correct Answer Feedback: Correct because the cost of the fiduciary call must equal the cost of the synthetic protective put, giving us what is referred to as put–call–forward parity: F (T)/(1 + r)T + p = c + X/(1 + r)T where c is the price of the European call, p is 0 0 0 0 0 the price of the European Put, F is the forward price of the underlying, X is the exercise 0 price, r is the risk-free rate and T is time. Therefore the price of a 9-month fiduciary call is equal to the price of a 9-month synthetic protective put and can be calculated by c + 0 X/(1 + r)T = \$3.25 + \$9.75/(1 + 0.015)9/12 = \$3.25 + \$9.64 = \$12.89.

Correct Answer Feedback: Correct because the cost of the fiduciary call must equal the cost of the synthetic protective put, giving us what is referred to as put–call–forward parity: $F_0(T)/(1+r)^T + p_0 = c_0 + X/(1+r)^T$ where c_0 is the price of the European call, p_0 is the price of the European Put, F_0 is the forward price of the underlying, X is the exercise price, r is the risk-free rate and T is time. Therefore the price of a 9-month fiduciary call is equal to the price of a 9-month synthetic protective put and can be calculated by $c_0 + X/(1+r)^T$ = \$3.25 + \$9.75/$(1 + 0.015)^{9/12}$ = \$3.25 + \$9.64 = \$12.89.

Figure 22: Source equation for question 29, option A

X is the exercise 0 price, r is the risk-free rate and T is time. Therefore the price of a 9-month fiduciary call is not equal to c + X or $3.25 + $9.75 = $13.00. 0

Incorrect Answer Feedback: Incorrect because the cost of the fiduciary call must equal the cost of the synthetic protective put, giving us what is referred to as put–call–forward parity: $F_0(T)/(1 + r)^T + p_0 = c_0 + X/(1 + r)^T$ where c_0 is the price of the European call, p_0 is the price of the European Put, F_0 is the forward price of the underlying, X is the exercise price, r is the risk-free rate and T is time. Therefore the price of a 9-month fiduciary call is not equal to $c_0 + X$ or $3.25 + $9.75 = $13.00.

Figure 23: Source equation for question 29, option B

C. $13.11. - Incorrect

Incorrect Answer Feedback: Incorrect because the cost of the fiduciary call must equal the cost of the synthetic protective put, giving us what is referred to as put–call–forward parity: F (T)/(1 + r)T + p = c + X/(1 + r)T where c is the price of the European call, p is 0 0 0 0 0 the price of the European Put, F is the forward price of the underlying, X is the exercise 0 price, r is the risk-free rate and T is time. Therefore the price of a 9-month fiduciary call is not equal to c + X × (1 + r)T or $3.25 + $9.75 × (1 + 0.015)9/12 = $3.25 + $9.86 = $13.11 0

Incorrect Answer Feedback: Incorrect because the cost of the fiduciary call must equal the cost of the synthetic protective put, giving us what is referred to as put–call–forward parity: $F_0(T)/(1 + r)^T + p_0 = c_0 + X/(1 + r)^T$ where c_0 is the price of the European call, p_0 is the price of the European Put, F_0 is the forward price of the underlying, X is the exercise price, r is the risk-free rate and T is time. Therefore the price of a 9-month fiduciary call is not equal to $c_0 + X \times (1 + r)^T$ or $\$3.25 + \$9.75 \times (1 + 0.015)^{9/12} = \$3.25 + \$9.86 = \$13.11$

Figure 24: Source equation for question 29, option C

Question 30

An investor gathers the following information about a European call option with 9 months until expiration: Current stock price $112

Current stock price	$112
Strike price of option	$115
Current call option price	$4
1-year risk free rate	2.5%

According to put–call parity, the current price of a put option with the same maturity and strike price is closest to:

Correct answer: C

A. $3.11. - Incorrect

Incorrect Answer Feedback: Incorrect because the current value of the put option is $4.89, according to put-call parity formula: p – c = X / (1 + r)T – S where p is the put 0 0 0, 0 price, c is the call price, X is the strike price, r is the risk free rate with T time to 0 expiration and S is the current stock price. p = $115 / (1 + 0.025)0.75 – $112 + $4 = 0, 0 $4.89. Switching p with c in the formula and solving for c will result in the wrong 0 0 0 answer $3.11 as c = p -X / (1 + r)T +S or c = $4 -$115 / (1 + 0.025)0.75 + $112 = $3.11 0 0 - 0 0

B. $4.20. - Incorrect

Incorrect Answer Feedback: Incorrect because the current value of the put option is $4.89, according to put-call parity formula: p – c = X / (1 + r)T – S where p is the put 0 0 0, 0 price, c is the call price, X is the strike price, r is

Incorrect Answer Feedback: Incorrect because the current value of the put option is $4.89, according to put-call parity formula: $p_0 - c_0 = X/(1+r)^T - S_0$, where p_0 is the put price, c_0 is the call price, X is the strike price, r is the risk free rate with T time to expiration and S_0 is the current stock price. $p_0 = \$115 / (1 + 0.025)^{0.75} - \$112 + \$4 = \$4.89$. Switching p_0 with c_0 in the formula and solving for c_0 will result in the wrong answer $3.11 as $c_0 = p_0 - X/(1+r)^T + S_0$ or $c_0 = \$4 - \$115 / (1 + 0.025)^{0.75} + \$112 = \$3.11$

Figure 25: Source equation for question 30, option A

the risk free rate with T time to 0 expiration and S is the current stock price. p = $115 / (1 + 0.025)0.75 – $112 + $4 = 0, 0 1 $4.89. Assuming T = 1 will result in the wrong answer $4.20 as $115 / (1 + 0.025) – $112 + $4 = $4.20.

Incorrect Answer Feedback: Incorrect because the current value of the put option is $4.89, according to put-call parity formula: $p_0 - c_0 = X/(1+r)^T - S_0$, where p_0 is the put price, c_0 is the call price, X is the strike price, r is the risk free rate with T time to expiration and S_0 is the current stock price. $p_0 = \$115 / (1 + 0.025)^{0.75} - \$112 + \$4 = \$4.89$. Assuming T = 1 will result in the wrong answer $4.20 as $115 / (1 + 0.025)^1 – $112 + $4 = $4.20.

Figure 26: Source equation for question 30, option B

C. $4.89. - Correct

Correct Answer Feedback: Correct because the current value of the put option is $4.89, according to put-call parity formula: p – c = X / (1 + r)T – S where p is the put 0 0 0, 0 price, c is the call price, X is the strike price, r is the risk free rate with T time to 0 0.75 expiration and S is the current stock price. p = $115 / (1 + 0.025) – $112 + $4 = 0, 0 $4.89.

Correct Answer Feedback: Correct because the current value of the put option is $4.89, according to put-call parity formula: $p_0 - c_0 = X/(1+r)^T - S_0$, where p_0 is the put price, c_0 is the call price, X is the strike price, r is the risk free rate with T time to expiration and S_0 is the current stock price. $p_0 = \$115 / (1 + 0.025)^{0.75} - \$112 + \$4 = \$4.89$.

ıo answer)

Figure 27: Source equation for question 30, option C

Question 31

An analyst gathers the following information about spot rates: Time Period Spot Rate 1 year 2% 2 year 3% 3 year 4% Using spot rates, the price of a 3-year, 6% annual coupon bond is closest to:

Correct answer: C

A. 105.38. - Incorrect

Incorrect Answer Feedback: Incorrect because it incorrectly uses “3” as the exponent in the denominator of each component of the calculation: PV = 6/ (1 + 0.02)3 + 6 / (1 + 0.03)3 + (6 + 100) / (1 + 0.04)3 = 105.38.

B. 105.55. - Incorrect

Incorrect Answer Feedback: Incorrect because it incorrectly uses the 3-year spot rate in the denominator of each component of the calculation: PV = 6/ (1 + 0.04)1 + 6 / (1 + 0.04)2 + (6 + 100) / (1 + 0.04)3 = 105.55.

Incorrect Answer Feedback: Incorrect because it incorrectly uses “3” as the exponent in the denominator of each component of the calculation:

Figure 28: Source equation for question 31, option A

$$PV = 6/ (1 + 0.02)^3 + 6 / (1 + 0.03)^3 + (6 + 100) / (1 + 0.04)^3 = 105.38.$$

Figure 29: Source equation for question 31, option A

C. 105.77. - Correct

Correct Answer Feedback: Correct because the general formula for calculating a bond price given the sequence of spot rates: PV = PMT / (1 + Z)1 + PMT / (1 + Z)2 + … + (PMT + FV) / (1 + Z)N 1 2 N Where PV = present value, or the price of the bond PMT = coupon payment per period FV = future value paid at maturity, or the par value of the bond Z = spot rate, or the zero-coupon yield, or zero rate, for Period 1 1 Z = spot rate, or the zero-coupon yield, or zero rate, for Period 2 2 Z = spot rate, or the zero-coupon yield, or zero rate, for Period N: N PV = 6/ (1 + 0.02)1 + 6 / (1 + 0.03)2 + (6 + 100) / (1 + 0.04)3 = 105.77.

Question 32

If yields decrease by 0.75%, the convexity adjustment for a bond with an annual convexity of 400 is:

Correct answer: A

A. 0.01125. - Correct

Correct Answer Feedback: Correct because the “second order” effect, is the convexity adjustment. The convexity adjustment is the annual convexity statistic, AnnConvexity, times one-half, multiplied by the change in the yield-to-maturity squared. Convexity adjustment = (0.5 × AnnConvexity × ΔYield2) 2) = (0.5 × 400 × 0.0075 = 0.01125.

B. 0.02250. - Incorrect

Incorrect Answer Feedback: Incorrect because the 0.5 is missing in the formula: Convexity adjustment ≠ (400 × 0.00752) = 0.02250

C. 1.50000. - Incorrect

Incorrect Answer Feedback: Incorrect because the change in yield is not squared: Convexity adjustment ≠ (0.5 × 400 × 0.0075) = 1.50000.

Question 33

Compared to a defined benefit plan’s investment time horizon, a bank’s investment time horizon is most likely:

Correct answer: A

A. shorter. - Correct

Correct Answer Feedback: Correct because the defined benefit plan may have an indefinitely long time horizon if new plan members are being admitted. As for banks, the time horizon is short term.

Incorrect Answer Feedback: Incorrect because it incorrectly uses the 3-year spot rate in the denominator of each component of the calculation:

$$PV = 6/ (1 + 0.04)^1 + 6 / (1 + 0.04)^2 + (6 + 100) / (1 + 0.04)^3 = 105.55.$$

Figure 30: Source equation for question 31, option B

Correct Answer Feedback: Correct because the general formula for calculating a bond price given the sequence

of spot rates: $PV = PMT / (1 + Z_1)^1 + PMT / (1 + Z_2)^2 + \ldots + (PMT + FV) / (1 + Z_N)^N$

Where

PV = present value, or the price of the bond

PMT = coupon payment per period

FV = future value paid at maturity, or the par value of the bond

Z_1 = spot rate, or the zero-coupon yield, or zero rate, for Period 1

Z_2 = spot rate, or the zero-coupon yield, or zero rate, for Period 2

Z_N = spot rate, or the zero-coupon yield, or zero rate, for Period N:

$PV = 6/ (1 + 0.02)^1 + 6 / (1 + 0.03)^2 + (6 + 100) / (1 + 0.04)^3 = 105.77.$

ıo answer)

Figure 31: Source equation for question 31, option C

Correct Answer Feedback: Correct because the "second order" effect, is the convexity adjustment. The convexity adjustment is the annual convexity statistic, AnnConvexity, times one-half, multiplied by the change in the yield-to-maturity squared.

Convexity adjustment = $(0.5 \times \text{AnnConvexity} \times \Delta\text{Yield}^2)$

= $(0.5 \times 400 \times 0.0075^{2)} = 0.01125.$

Figure 32: Source equation for question 32, option A

Incorrect Answer Feedback: Incorrect because the 0.5 is missing in the formula:

Convexity adjustment ≠ $(400 \times 0.0075^{2)} = 0.02250$

Figure 33: Source equation for question 32, option B

Figure 34: Source equation for question 32, option B

Incorrect Answer Feedback: Incorrect because the change in yield is not squared:

Convexity adjustment ≠ $(0.5 \times 400 \times 0.0075) = 1.50000.$

ıo answer)

Figure 35: Source equation for question 32, option C

B. the same. - Incorrect

Incorrect Answer Feedback: Incorrect because banks typically have much shorter investment time horizons than defined benefit plans.

C. longer. - Incorrect

Incorrect Answer Feedback: Incorrect because banks typically have much shorter investment time horizons than defined benefit plans.

Question 34

Management fees for private equity funds are most likely based on:

Correct answer: B

A. invested capital. - Incorrect

Incorrect Answer Feedback: Incorrect because private equity funds raise committed capital and draw down on those commitments, generally over three to five years, when they have a specific investment to make. Note that the management fee is typically based on committed capital, not invested capital; the committed-capital basis for management fees is an important distinction from hedge funds, whose management fees are based on assets under management (AUM).

B. committed capital - Correct

Correct Answer Feedback: Correct because private equity funds raise committed capital and draw down on those commitments, generally over three to five years, when they have a specific investment to make. Note that the management fee is typically based on committed capital, not invested capital; the committed-capital basis for management fees is an important distinction from hedge funds, whose management fees are based on assets under management (AUM).

C. assets under management. - Incorrect

Incorrect Answer Feedback: Incorrect because private equity funds raise committed capital and draw down on those commitments, generally over three to five years, when they have a specific investment to make. Note that the management fee is typically based on committed capital, not invested capital; the committed-capital basis for management fees is an important distinction from hedge funds, whose management fees are based on assets under management (AUM).

Question 35

A portfolio manager analyzes a market and discovers that it is not possible to achieve consistent and superior risk-adjusted returns, net of all expenses. This market is most likely characterized by:

Correct answer: B

A. persistent anomalies. - Incorrect

Incorrect Answer Feedback: Incorrect because in an informationally efficient market, consistent and superior risk-adjusted returns (net of all expenses) are not achievable. In such a situation, persistent anomalies are unlikely.

B. informational efficiency. - Correct

Correct Answer Feedback: Correct because in an informationally efficient market, consistent and superior risk-adjusted returns (net of all expenses) are not achievable.

C. restrictions on short selling. - Incorrect

Incorrect Answer Feedback: Incorrect because some market experts argue that restrictions on short selling limit arbitrage trading, which impedes market efficiency.

Question 36

Which of the following statements about the use of derivative instruments is most accurate?

Correct answer: B

A. Small exposures to an underlying can be created with a large cash outlay - Incorrect

Incorrect Answer Feedback: Incorrect because the opposite is true: Market participants may create large (not small) exposures to an underlying with a relatively small cash outlay.

B. Financial market exposure associated with commercial transactions can be offset by issuers - Correct

Correct Answer Feedback: Correct because derivative markets expand the set of opportunities available to market participants to create or modify exposure to an underlying in several ways, including allowing Issuers to offset the financial market exposure associated with a commercial transaction.

C. Higher transaction costs are typically incurred compared to underlying spot market transactions - Incorrect

Incorrect Answer Feedback: Incorrect because the opposite is true: Derivatives typically have lower (not higher) transaction costs and are often more liquid than underlying spot market transactions.

Question 37

Which of the following cognitive errors is best classified as a belief perseverance bias?

Correct answer: B

A. Availability bias - Incorrect

Incorrect Answer Feedback: Incorrect because it is a processing error, not a belief perseverance bias. The processing errors discussed are anchoring and adjustment, mental accounting, framing, and availability.

B. Confirmation bias - Correct

Correct Answer Feedback: Correct because the belief perseverance biases discussed are conservatism, confirmation, representativeness, illusion of control, and hindsight. Confirmation bias refers to the tendency to look for and notice what confirms prior beliefs and to ignore or undervalue whatever contradicts them.

C. Mental accounting bias - Incorrect

Incorrect Answer Feedback: Incorrect because it is a processing error, not a belief perseverance bias. The processing errors discussed are anchoring and adjustment, mental accounting, framing, and availability.

Question 38

The redemption value of a US Treasury Inflation-Protected Securities (TIPS) issue at maturity is most likely:

Correct answer: B

A. equal to the initial par value. - Incorrect

Incorrect Answer Feedback: Incorrect because investors receive the greater of inflation-adjusted or par at maturity. TIPS, which have a periodic principal adjustment based on a price index, are an example of a capital-indexed bond. Note that while deflation causes the principal to decline over time, investors usually receive the greater of the inflation-adjusted principal or par at maturity.

B. the greater of the inflation-adjusted principal or initial par value. - Correct

Correct Answer Feedback: Correct because TIPS, which have a periodic principal adjustment based on a price index, are an example of a capital-indexed bond. Note that while deflation causes the principal to decline over time, investors usually receive the greater of the inflation-adjusted principal or par at maturity.

C. above or below the initial par value depending on actual inflation during the term. - Incorrect

Incorrect Answer Feedback: Incorrect because investors can never receive below par. TIPS, which have a periodic principal adjustment based on a price index, are an example of a capital-indexed bond. Note that while deflation causes the principal to decline over time, investors usually receive the greater of the inflation-adjusted principal or par at maturity.

Question 39

Tactical asset allocation is best described as adding value by:

Correct answer: B

A. subdividing portfolio risk over the sources of investment return. - Incorrect

Incorrect Answer Feedback: Incorrect because it describes risk budgeting, not tactical asset allocation. Risk budgeting is the process of deciding on the amount of risk to assume in a portfolio (the overall risk budget), and subdividing that risk over the sources of investment return (e.g., strategic asset allocation, tactical asset allocation, and security selection).

B. deliberately deviating from policy exposures to systematic risk factors. - Correct

Correct Answer Feedback: Correct because tactical asset allocation is the decision to deliberately deviate from the policy exposures to systematic risk factors (i.e., the policy weights of asset classes) with the intent to add value based on forecasts of the near-term returns of those asset classes.

C. combining long-term capital market expectations with investor objectives and constraints. - Incorrect

Incorrect Answer Feedback: Incorrect because it describes strategic asset allocation, not tactical asset allocation. A strategic asset allocation results from combining the constraints and objectives articulated in the IPS and long-term capital market expectations regarding the asset classes.

Question 40

In a one-period binomial model, the risk-neutral probabilities of upward and downward price movements of an underlying asset are determined by:

Correct answer: A

A. the risk-free rate. - Correct

Correct Answer Feedback: Correct because in a one-period binomial model, the risk- neutral probabilities are determined only by the risk-free rate over the life of the option and the underlying asset's volatility (as measured by the up and down gross returns, R and R). u d

B. investors' risk aversion. - Incorrect

Incorrect Answer Feedback: Incorrect because of the ability to construct a perfect hedge of the option using the underlying asset, an option's price is independent of investors' risk aversion and the probability of the underlying price moving up (or down).

C. the probabilities of the underlying price moving up and down. - Incorrect

Incorrect Answer Feedback: Incorrect because of the ability to construct a perfect hedge of the option using the underlying asset, an option's price is independent of investors' risk aversion and the probability of the underlying price moving up (or down).

Question 41

An investor buys a call for $24.70 that has a strike price of $650. If the value at expiration for this call is $47.60, the price of the underlying at expiration is closest to:

Correct answer: C

A. $602.40. - Incorrect

Incorrect Answer Feedback: Incorrect because $602.40 represents the value or “payoff to the put buyer” at expiration, which is p = Max(0, X – S), where X is the exercise T T price and S is the price of the underlying at expiration. Given the information in the T stem we get $47.60 = Max(0,$650 – S) which yields S = $650 – $47.60 = $602.40. T T By contrast, the value or “payoff to the call buyer” at expiration is c = Max(0,S – T T X). Given the information in the stem we get $47.60 = Max(0,S – $650). Hence, S = T T 650 + 47.60 = 697.60, not $602.40.

Incorrect Answer Feedback: Incorrect because $602.40 represents the value or "payoff to the put buyer" at expiration, which is p_T = Max(0, X – S_T), where X is the exercise price and S_T is the price of the underlying at expiration. Given the information in the stem we get $47.60 = Max(0,$650 – S_T) which yields S_T = $650 – $47.60 = $602.40. By contrast, the value or "payoff to the call buyer" at expiration is c_T = Max(0,S_T – X). Given the information in the stem we get $47.60 = Max(0,$S_T$ – $650). Hence, S_T = 650 + 47.60 = 697.60, not $602.40.

Figure 36: Source equation for question 41, option A

B. $672.90. - Incorrect

Incorrect Answer Feedback: Incorrect because $672.90 simply represents the difference between $697.60 and $24.70. By contrast, the value or “payoff to the call buyer” at expiration is c = Max(0,S – X) where X is the strike price, S is the price of T T T the underlying at expiration. Given the information in the stem we get $47.60 = Max(0,S – $650). Hence, S =$ 650 + $47.60 = $697.60, not $672.90. T T

Incorrect Answer Feedback: Incorrect because $672.90 simply represents the difference between $697.60 and $24.70. By contrast, the value or "payoff to the call buyer" at expiration is c_T = Max(0,S_T – X) where X is the strike price, S_T is the price of the underlying at expiration. Given the information in the stem we get $47.60 = Max(0,$S_T$ – $650). Hence, S_T =$ 650 + $47.60 = $697.60, not $672.90.

Figure 37: Source equation for question 41, option B

C. $697.60. - Correct

Correct Answer Feedback: Correct because the value or “payoff to the call buyer” at expiration is c = Max(0,S – X) where X is the strike price, S is the price of the T T T underlying at expiration. Given the information in the stem we get $47.60 = Max(0,S – T $650). Hence, S =$ 650 + $47.60 = $697.60. T

Correct Answer Feedback: Correct because the value or "payoff to the call buyer" at expiration is c_T = Max(0,S_T – X) where X is the strike price, S_T is the price of the underlying at expiration. Given the information in the stem we get $47.60 = Max(0,$S_T$ – $650). Hence, S_T =$ 650 + $47.60 = $697.60.

ıo answer)

Figure 38: Source equation for question 41, option C

Question 42

An annual-pay bond with a £1,000 face value has a coupon rate of 4% and matures in three years. If the yield to maturity is 3%, the current value of the bond is closest to:

Correct answer: B

A. £972. - Incorrect

Incorrect Answer Feedback: Incorrect because it changes the payment amount to semiannual but leaves the time period and discount rate as annual. FV = £1,000 I = 3 N = 3 PMT = 20 PV = £972 rounded

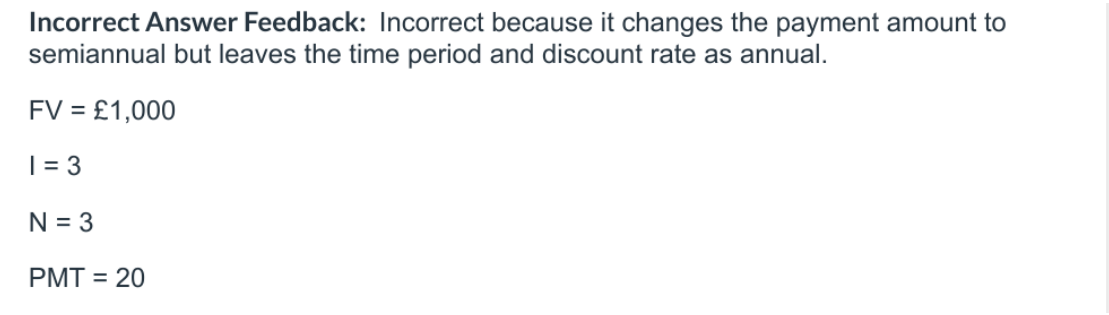


Figure 39: Source equation for question 42, option A

Figure 40: Source equation for question 42, option A

B. £1,028. - Correct

Correct Answer Feedback: Correct because FV = £1,000 I = 3 N = 3 PMT = 40 PV = £1,028 Or PV = 40/1.03 + 40/(1.03)2 + 1,040/(1.03)3 = £1,028.30, rounded to £1,028

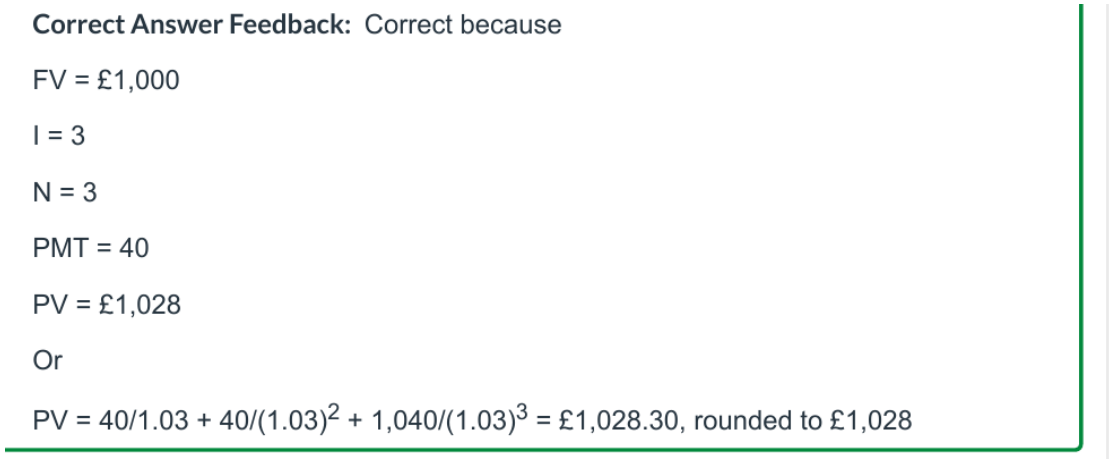


Figure 41: Source equation for question 42, option B

C. £1,142. - Incorrect

Incorrect Answer Feedback: Incorrect because it changes the discount rate and time period to semiannual and leaves the payment as annual. FV = £1,000 I = 1.5 N = 6 PMT = 40 PV = £1,142

Question 43

With respect to the absolute priority of claims in bankruptcy, which of the following most likely has the highest seniority ranking?

Correct answer: A

A. Third lien debt - Correct

Correct Answer Feedback: Correct because in the event of default, unsecured debtholders' claims rank below (i.e., get paid after) those of secured creditors under what's known as the priority of claims. Within secured debt there can also be second lien, or even third lien, secured debt, which, as the name implies, has a secured interest

Incorrect Answer Feedback: Incorrect because it changes the discount rate and time period to semiannual and leaves the payment as annual.

FV = £1,000

I = 1.5

N = 6

PMT = 40

PV = £1,142

ıo answer)

Figure 42: Source equation for question 42, option C

in the pledged assets but ranks below first lien debt in both collateral protection and priority of payment. Other, lower-ranked debt includes subordinated debt and junior subordinated debt.

B. Subordinated debt - Incorrect

Incorrect Answer Feedback: Incorrect because as specified in the justification for the correct response, subordinated debt ranks below third lien debt (secured debt). C. Senior unsecured debt

C. Senior unsecured debt - Incorrect

Incorrect Answer Feedback: Incorrect because as specified in the justification for the correct response, senior unsecured debt ranks below third lien debt (secured debt).

Question 44

Compared with its market-value-weighted counterpart, a fundamental-weighted index is least likely to have a:

Correct answer: C

A. value tilt. - Incorrect

Incorrect Answer Feedback: Incorrect because fundamental weighting leads to a value tilt because the ratios of book value, earnings, dividends, etc., to market value of the firms in a fundamental index tend to be larger than those of the firms in its market- capitalization-weighted counterpart.

B. contrarian effect. - Incorrect

Incorrect Answer Feedback: Incorrect because the fundamental indexes generally have a contrarian effect in that the portfolio weights will shift away from securities that have increased in relative value whenever the portfolio is rebalanced.

C. momentum effect. - Correct

Correct Answer Feedback: Correct because the momentum effect is a characteristic of a market-capitalization-weighted index, not a fundamental index.

Question 45

In contrast to risk tolerance, risk budgeting focuses more on:

Correct answer: A

A. how risk is taken. - Correct

Correct Answer Feedback: Correct because whereas risk tolerance focuses on the appetite for risk and what is and is not acceptable, risk budgeting has a more specific focus on how that risk is taken.

B. the appetite for risk. - Incorrect

Incorrect Answer Feedback: Incorrect because whereas risk tolerance focuses on the appetite for risk and what is and is not acceptable, risk budgeting has a more specific focus on how that risk is taken.

C. what is and is not acceptable risk. - Incorrect

Incorrect Answer Feedback: Incorrect because whereas risk tolerance focuses on the appetite for risk and what is and is not acceptable, risk budgeting has a more specific focus on how that risk is taken.

Question 46

Which of the following statements is most accurate? Cryptocurrencies:

Correct answer: C

A. exhibit low volatility. - Incorrect

Incorrect Answer Feedback: Incorrect because it is important to note that many cryptocurrencies have experienced high levels of price volatility. A lack of clear fundamentals underlying these currencies has contributed to their volatility.

B. have no limits on the total amount of currency that may be issued. - Incorrect

Incorrect Answer Feedback: Incorrect because many cryptocurrencies have a self- imposed limit on the total amount of currency they may issue.

C. allow transactions between parties without the need for an intermediary. - Correct

Correct Answer Feedback: Correct because a cryptocurrency, also known as a digital currency, operates as electronic currency and allows near-real-time transactions between parties without the need for an intermediary, such as a bank.

Question 47

The duration statistic that best measures the interest rate risk of a bond that does not have a well-defined internal rate of return is:

Correct answer: A

A. effective duration. - Correct

Correct Answer Feedback: Correct because effective duration is essential to the measurement of the interest rate risk of a complex bond, such as a bond that contains an embedded call option a callable bond does not have a well-defined internal rate of return (yield-to-maturity). Therefore, yield duration statistics, such as modified and Macaulay durations, do not apply; effective duration is the appropriate duration measure.

B. modified duration. - Incorrect

Incorrect Answer Feedback: Incorrect because effective duration is essential to the measurement of the interest rate risk of a complex bond, such as a bond that contains an embedded call option a callable bond does not have a well-defined internal rate of return (yield-to-maturity). Therefore, yield duration statistics, such as modified and Macaulay durations, do not apply; effective duration is the appropriate duration measure.

C. Macaulay duration. - Incorrect

Incorrect Answer Feedback: Incorrect because effective duration is essential to the measurement of the interest rate risk of a complex bond, such as a bond that contains an embedded call option a callable bond does not have a well-defined internal rate of return (yield-to-maturity). Therefore, yield duration statistics, such as modified and Macaulay durations, do not apply; effective duration is the appropriate duration measure.

Question 48

An analyst gathers the following information about three well-diversified equity portfolios:

Portfolio	Mean return	Beta
1	9.2%	1.2
2	12.4%	1.4
3	15.0%	2.3

Given a risk-free rate of 1.8%, the portfolio with the highest Treynor ratio is:

Correct answer: B

A. Portfolio 1. - Incorrect

Incorrect Answer Feedback: Incorrect because the Treynor ratio of Portfolio 1 ranks above Portfolio 3, but below Portfolio 2. TR = (R − R) / β 1 1 f 1 = (9.2% − 1.8%) / 1.2 = 0.0617

Incorrect Answer Feedback: Incorrect because the Treynor ratio of Portfolio 1 ranks above Portfolio 3, but below Portfolio 2.

$TR_1 = (R_1 - R_f) / \beta_1$

$= (9.2\% - 1.8\%) / 1.2$

$= 0.0617$

Figure 43: Source equation for question 48, option A

B. Portfolio 2. - Correct

Correct Answer Feedback: Correct because the Treynor Ratio of Portfolio 2 is the highest of the three portfolios. TR = (R − R) / β 2 2 f 2 = (12.4% − 1.8%) / 1.4 = 0.0757

Portfolio	Calculation	Treynor Ratio
1	(9.2% − 1.8%) / 1.2	0.0617
2	(12.4% − 1.8%) / 1.4	0.0757
3	(15.0% − 1.8%) / 2.3	0.0574

The Treynor ratio is a simple extension of the Sharpe ratio and resolves the Sharpe ratio's first limitation by substituting beta (systematic risk) for total risk. The portfolio with the highest Treynor ratio will be the best performing based on that measure.

C. Portfolio 3. - Incorrect

Incorrect Answer Feedback: Incorrect because Portfolio 3 has the lowest Treynor ratio of the three portfolios. TR = (R − R) / β 3 3 f 3 = (15.0% − 1.8%) / 2.3 = 0.0574

Correct Answer Feedback: Correct because the Treynor Ratio of Portfolio 2 is the highest of the three portfolios.

$TR_2 = (R_2 - R_f) / \beta_2$

$= (12.4\% - 1.8\%) / 1.4$

$= 0.0757$

Portfolio	Calculation	Treynor Ratio

Figure 44: Source equation for question 48, option B

2	(12.4% − 1.8%) / 1.4	0.0757
3	(15.0% − 1.8%) / 2.3	0.0574

The Treynor ratio is a simple extension of the Sharpe ratio and resolves the Sharpe ratio's first limitation by substituting beta (systematic risk) for total risk. The portfolio with the highest Treynor ratio will be the best performing based on that measure.

Figure 45: Source equation for question 48, option B

Incorrect Answer Feedback: Incorrect because Portfolio 3 has the lowest Treynor ratio of the three portfolios.

$TR_3 = (R_3 - R_f) / \beta_3$

$= (15.0\% - 1.8\%) / 2.3$

$= 0.0574$

ıo answer)

Figure 46: Source equation for question 48, option C

Question 49

The modified duration of an option-free corporate bond best estimates the percentage price change due to changes in the:

Correct answer: B

A. horizon yield. - Incorrect

Incorrect Answer Feedback: Incorrect because a horizon yield is the internal rate of return between the total return (the sum of reinvested coupon payments and the sale price or redemption amount) and the purchase price of the bond. The horizon yield on a bond investment is the annualized holding-period rate of return. The horizon yield is a historical measure of yield for an investor's holding period, and not an estimate of future bond price changes.

B. yield-to-maturity. - Correct

Correct Answer Feedback: Correct because modified duration provides an estimate of the percentage price change for a bond given a change in its yield-to-maturity.

C. benchmark yield curve. - Incorrect

Incorrect Answer Feedback: Incorrect because modified duration is a yield duration statistic in that it measures interest rate risk in terms of a change in the bond's own yield- to-maturity whereas, effective duration of a bond is the sensitivity of the bond's price to a change in a benchmark yield curve.

Question 50

An analyst gathers the following information about a portfolio:

	Portfolio Weight	Standard Deviation
Asset 1	25%	12%
Asset 2	75%	16%

If the correlation coefficient between the two assets is 0.75, the standard deviation of the portfolio is closest to:

Correct answer: B

A. 12.37%. - Incorrect

Incorrect Answer Feedback: Incorrect because it omits the third term in the 2 2 2 formula in calculating the standard deviation: [(0.25 × 0.12) + (0.75 × 0.162)]0.5 ≈ 0.1237 = 12.37%.

Incorrect Answer Feedback: Incorrect because it omits the third term in the formula in calculating the standard deviation: $[(0.25^2 \times 0.12^2) + (0.75^2 \times 0.16^2)]^{0.5} \approx 0.1237 = 12.37\%$.

Figure 47: Source equation for question 50, option A

B. 14.39%. - Correct

Correct Answer Feedback: Correct because the formula for calculating the standard deviation of a two stock portfolio, σ , is: p 2 2 2 2 1/2 σ = [w σ + w σ + 2w w ρ σ σ] p 1 1 2 2 1 2 12 1 2 where: w = weight of each stock σ = standard deviation ρ = the correlation of returns between Stock 1 and Stock 2 12 Plugging in the numbers, σ = p 2 2 2 2 0.5 [(0.25 × 0.12) + (0.75 × 0.16) + (2 × 0.25 × 0.75 × 0.12 × 0.16 × 0.75)] ≈ 0.1439 = 14.39%

Correct Answer Feedback: Correct because the formula for calculating the standard deviation of a two stock portfolio, σ_p, is:

$$\sigma_p = [w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{12}\sigma_1\sigma_2]^{1/2}$$

where:

w = weight of each stock

σ = standard deviation

ρ_{12} = the correlation of returns between Stock 1 and Stock 2

Plugging in the numbers, σ_p =

$[(0.25^2 \times 0.12^2) + (0.75^2 \times 0.16^2) + (2 \times 0.25 \times 0.75 \times 0.12 \times 0.16 \times 0.75)]^{0.5} \approx 0.1439 = 14.39\%$

Figure 48: Source equation for question 50, option B

C. 15.00%. - Incorrect

Incorrect Answer Feedback: Incorrect because it omits the correlation 2 2 2 coefficient in calculating the standard deviation: [(0.25 × 0.12) + (0.75 × 0.162) + (2 × 0.25 × 0.75 × 0.12 × 0.16)]0.5 = 0.15 = 15%, which is the weighted average standard deviation.

Incorrect Answer Feedback: Incorrect because it omits the correlation coefficient in calculating the standard deviation: $[(0.25^2 \times 0.12^2) + (0.75^2 \times 0.16^2) + (2 \times 0.25 \times 0.75 \times 0.12 \times 0.16)]^{0.5} = 0.15 = 15\%$, which is the weighted average standard deviation.

Figure 49: Source equation for question 50, option C

Question 51

Two investors have utility functions that differ only with regard to the coefficient of risk aversion. Relative to the investor with a higher coefficient of risk aversion, the optimal portfolio for the investor with a lower coefficient of risk aversion will most likely have:

Correct answer: C

A. a lower level of risk and return. - Incorrect

Incorrect Answer Feedback: Incorrect. For a high coefficient of risk aversion, the investor will seek a lot of return for a bit of extra risk and will have an optimal portfolio that is tangential to the capital allocation line at a lower level of risk and return relative to the investor with a lower coefficient of risk aversion.

B. the same level of risk and return. - Incorrect

Incorrect Answer Feedback: Incorrect. Only investors with identical coefficients of risk aversion would select the same optimal portfolio.

C. a higher level of risk and return. - Correct

Correct Answer Feedback: Correct. A less risk-averse investor's highest utility curve, given the lower coefficient of risk aversion, is likely to touch the capital allocation line at a point that would represent a portfolio with higher risk and more expected return.

Question 52

Which of the following involves a private equity firm introducing leverage to its portfolio company and paying itself a dividend out of the new capital structure?

Correct answer: B

A. Venture debt - Incorrect

Incorrect Answer Feedback: Incorrect because it's one of the primary methods. We can organize the primary methods of private debt investing into four categories: direct lending, mezzanine loans, venture debt, and distressed debt.

B. Recapitalization - Correct

Correct Answer Feedback: Correct because recapitalization via private equity describes the steps a firm takes to increase or introduce leverage to its portfolio company and pay itself a dividend out of the new capital structure.

C. Mezzanine loans - Incorrect

Incorrect Answer Feedback: Incorrect because it's one of the primary methods. We can organize the primary methods of private debt investing into four categories: direct lending, mezzanine loans, venture debt, and distressed debt.

Question 53

A European call option has an exercise price of $80 and an option premium of $12. The price of the underlying is $75. The intrinsic value of the option is:

Correct answer: A

A. $0. - Correct

Correct Answer Feedback: Correct because the value of the option at expiration is the greater of either zero or the underlying price at expiration minus the exercise price, which is typically written as C = Max(0,S – X) This formula is also sometimes referred T T to as the exercise value or intrinsic value. Intrinsic value of call option = Max(0,$75 – $80) = $0.

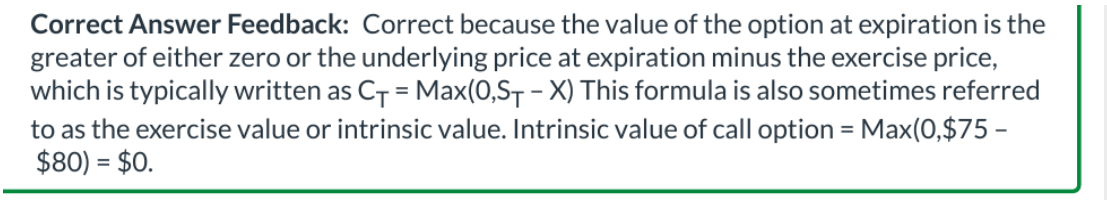


Figure 50: Source equation for question 53, option A

B. $5. - Incorrect

Incorrect Answer Feedback: Incorrect because this would be the intrinsic value if it were a put option. The value of a European put at expiration is the exercise value, which is the greater of zero or the exercise price minus the value of the underlying. Intrinsic value of the hypothetical put option = Max(0,$80 – $75) = $5.

Incorrect Answer Feedback: Incorrect because this would be the intrinsic value if it were a put option. The value of a European put at expiration is the exercise value, which is the greater of zero or the exercise price minus the value of the underlying. Intrinsic value of the hypothetical put option = Max(0,$80 – $75) = $5.

Figure 51: Source equation for question 53, option B

C. $7. - Incorrect

Incorrect Answer Feedback: Incorrect because this is the time value if it were a put option and not the intrinsic value of a call option. The time value of an option is the difference between the market price of the option and its intrinsic value. Time value of the put option = Market price –Intrinsic value = $12 – Max(0,$80 – $75) = $7.

Incorrect Answer Feedback: Incorrect because this is the time value if it were a put option and not the intrinsic value of a call option. The time value of an option is the difference between the market price of the option and its intrinsic value. Time value of the put option = Market price –Intrinsic value = $12 – Max(0,$80 – $75) = $7.

ıo answer)

Figure 52: Source equation for question 53, option C

Question 54

Which of the following most likely provides the greatest redemption flexibility?

Correct answer: C

A. Hedge funds - Incorrect

Incorrect Answer Feedback: Incorrect because alternative investments often have restrictions on redemptions (i.e., “lockups” and “gates”). Investors may be required to keep their money in the hedge fund for a minimum period (referred to as a lockup period) before they are allowed to make withdrawals or redeem shares. Furthermore, investors may be required to give notice of their intent to redeem; the notice period is typically 30–90 days. Therefore, a hedge fund does not offer the greatest redemption flexibility to an investor.

B. Private equity funds - Incorrect

Incorrect Answer Feedback: Incorrect because private equity investments generally involve an initial capital commitment, but actual capital flows often lag that commitment because capital “calls” are staggered over substantive periods of time. Private equity returns are frequently described in terms of the J-curve effect: a substantive initial capital commitment promise, followed by high initial fee drag, followed by the identification of longer-term growth or turnaround opportunities, and an eventual positive return (the investor hopes) when the staggered returns of the fund are realized. The component investments mature and are sold at various times, and the partnership finally closes—but usually only after six to eight years (often with two-year extensions allowed at the discretion of the general partner or after a vote of approval by a majority of limited partners). Therefore, a private equity fund does not offer the greatest redemption flexibility to an investor.

C. Funds of hedge funds - Correct

Correct Answer Feedback: Correct because funds of hedge funds may offer more redemption flexibility than is afforded to direct investors in hedge funds because of special redemption arrangements with the underlying hedge fund managers, the maintenance of added cash reserves, access to temporary bridge-loan financing, or the simple avoidance of less liquid hedge fund strategies.

Question 55

An equity portfolio manager is evaluating her sector allocation strategy for the upcoming year. She expects the global economy to experience a slowdown period for the next two years. Furthermore, she believes that

companies will be facing diminishing growth rates with respect to revenues and profits. On the basis of these beliefs, the portfolio manager will most likely overweight:

Correct answer: C

A. autos. - Incorrect

Incorrect Answer Feedback: Incorrect because the telecommunication sector exhibits less economic sensitivity, so it should be overweighted, not underweighted.

B. materials. - Incorrect

Incorrect Answer Feedback: Incorrect because the materials sector tends to exhibit a relatively high degree of economic sensitivity.

C. consumer staples. - Correct

Correct Answer Feedback: Correct because in periods of economic slowdowns, the manager would tend to overweight in non-cyclical companies, such as consumer staples.

Question 56

A company's depository receipts most likely trade:

Correct answer: A

A. like ordinary shares on the investors' local exchanges. - Correct

Correct Answer Feedback: Correct because a depository receipt (DR) is a security that trades like an ordinary share on an investor's local exchange and represents an economic interest in a foreign company. It allows the publicly listed shares of a foreign company to be traded on an exchange outside its domestic market.

B. in the local currency on the company's local exchange. - Incorrect

Incorrect Answer Feedback: Incorrect because a depository receipt (DR) is a security that trades like an ordinary share on an investor's local exchange and represents an economic interest in a foreign company. It allows the publicly listed shares of a foreign company to be traded on an exchange outside its domestic market. It trades in investors', not the company's, local currency.

C. in the company's currency on investors' local exchanges. - Incorrect

Incorrect Answer Feedback: Incorrect because depository receipts represent the equity of international companies and are traded on local exchanges and in the local currencies. With these securities, investors have to worry less about currency conversions (price quotations and dividend payments are in the investor's local currency), unfamiliar market practices, and differences in accounting standards.

Question 57

An investor opens a margin account with an initial deposit of $5,000. He then purchases 300 shares of a stock at $30 per share using margin. The account requires a maintenance margin of 30%. Ignoring commissions and interest, the price below which the investor will first receive a margin call is closest to:

Correct answer: A

A. $19.05. - Correct

Correct Answer Feedback: Correct because the price below which the investor will receive a margin call can be calculated using the equation for calculating maintenance margin. Maintenance margin = Equity ÷ Market value = Solving for Price, P = $19.05

Correct Answer Feedback: Correct because the price below which the investor will receive a margin call can be calculated using the equation for calculating maintenance

Figure 53: Source equation for question 57, option A

margin.

Maintenance margin = Equity ÷ Market value =

$$\frac{5{,}000+(P\times 300)-(30\times 300)}{P\times 300}=30\%$$

Solving for Price, P = $19.05

Figure 54: Source equation for question 57, option A

B. $23.08. - Incorrect

Incorrect Answer Feedback: Incorrect because this answer is derived by using the total purchase price of the stock instead of the margin loan. P = $23.08

Incorrect Answer Feedback: Incorrect because this answer is derived by using the total purchase price of the stock instead of the margin loan.

$$\frac{(P\times 300)-(30\times 300)}{P\times 300}=30\%$$

P = $23.08

Figure 55: Source equation for question 57, option B

C. $23.81. - Incorrect

Incorrect Answer Feedback: Incorrect because this answer is derived by subtracting the equity instead of the margin loan in the numerator. P = $23.81

Question 58

What is the most likely reason why arbitrage will not completely eliminate all pricing discrepancies for derivatives?

Correct answer: A

A. Transaction costs - Correct

Correct Answer Feedback: Correct. Transaction costs may render an arbitrage strategy unprofitable and can therefore prevent precise convergence of prices.

B. Inaccurate forecasts - Incorrect

Incorrect Answer Feedback: Incorrect. No forecasts are needed in implementing an arbitrage position.

C. Differences in risk aversion - Incorrect

Incorrect Answer Feedback: Incorrect. Differences in risk aversion are irrelevant for arbitrage because arbitrage transactions are riskless.

Question 59

Key rate duration measures the sensitivity of a bond's price to a change in:

Correct answer: C

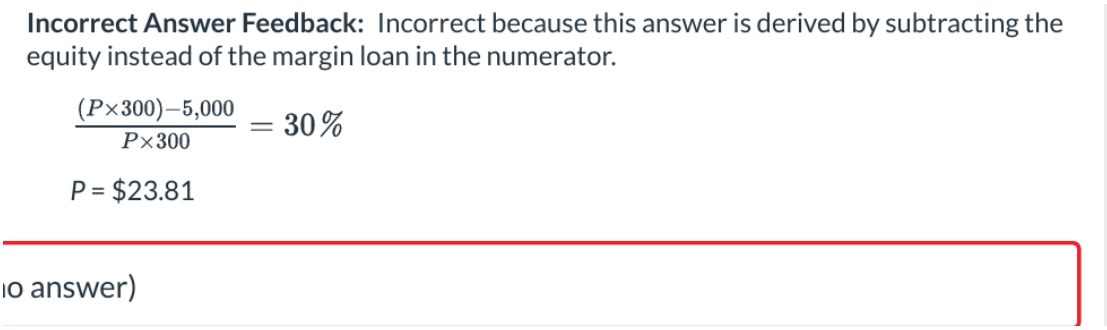


Figure 56: Source equation for question 57, option C

A. its yield to maturity. - Incorrect

Incorrect Answer Feedback: Incorrect because modified duration is a yield duration statistic in that it measures interest rate risk in terms of a change in the bond's own yield- to-maturity while key rate durations help identify 'shaping risk' for a bond—that is, a bond's sensitivity to changes in the shape of the benchmark yield curve.

B. its yield to first call. - Incorrect

Incorrect Answer Feedback: Incorrect because the duration of a callable bond is not the sensitivity of the bond price to a change in the yield-to-worst (i.e., the lowest of the yield- to-maturity, yield-to-first-call, yield-to-second-call, and so forth). The problem is that future cash flows are uncertain because they are contingent on future interest rates. The issuer's decision to call the bond depends on the ability to refinance the debt at a lower cost of funds. In brief, a callable bond does not have a well-defined internal rate of return (yield-to-maturity). Therefore, effective duration is the appropriate duration measure. Key rate duration is not used to measure the sensitivity of a callable bond's price to a change in the yield-to-first-call.

C. the shape of the benchmark yield curve. - Correct

Correct Answer Feedback: Correct because a key rate duration (or partial duration) is a measure of a bond's sensitivity to a change in the benchmark yield curve at a specific maturity segment. In contrast to effective duration, key rate durations help identify "shaping risk" for a bond—that is, a bond's sensitivity to changes in the shape of the benchmark yield curve.

Question 60

The sum of the coupon payments over a year divided by the flat price of a bond is best defined as:

Correct answer: C

A. true yield. - Incorrect

Incorrect Answer Feedback: Incorrect because the true yield-to-maturity is the internal rate of return on the cash flows using the actual calendar of weekends and bank holidays. True yield incorporates a bond's premium and discount to maturity unlike current yield.

B. simple yield. - Incorrect

Incorrect Answer Feedback: Incorrect because the simple yield on a bond is the sum of the coupon payments plus the straight-line amortized share of the gain or loss, divided by the flat price. The simple yield incorporates the premium and discount to maturity unlike current yield.

C. current yield. - Correct

Correct Answer Feedback: Correct because the current yield (also called the income or running yield) is the sum of the coupon payments received over the year divided by the flat price.

Question 61

An analyst is concerned that a bond might be downgraded one category by Standard & Poor's and become non-investment grade. The current rating of this bond is most likely:

Correct answer: C

A. A–. - Incorrect

Incorrect Answer Feedback: Incorrect because a bond rated A– can be downgraded one category (to BBB–) and still be an investment grade bond.

B. BB–. - Incorrect

Incorrect Answer Feedback: Incorrect because a bond rated BB– is already below investment grade before any downgrade.

C. BBB–. - Correct

Correct Answer Feedback: Correct because BBB– is the lowest rating for investment grade bonds. A one-category downgrade (from BBB– to BB–) would make the bond non- investment grade.

Question 62

A company that is required to raise equity capital to continue to operate as a going concern is most likely doing so to:

Correct answer: C

A. purchase long-lived assets. - Incorrect

Incorrect Answer Feedback: Incorrect because a firm is not required to use equity capital to purchase long-lived assets. Purchasing long-lived assets is a typical use of capital for companies that are raising capital to finance revenue generating activities, but this may not be the primary reason for a firm that is forced to raise capital to continue to operate as a going concern. B. fund capital expansion projects.

B. fund capital expansion projects. - Incorrect

Incorrect Answer Feedback: Incorrect because a firm is not required to use equity capital to fund capital expansion projects. Funding capital expansion projects is a typical use of capital for companies that are raising capital to finance revenue generating activities, but this may not be the primary reason for a firm that is forced to raise capital to continue to operate as a going concern.

C. improve capital adequacy ratios. - Correct

Correct Answer Feedback: Correct because in cases in which a company must raise capital to ensure it can continue to operate as a going concern, capital is most likely raised to fulfill regulatory requirements, improve capital adequacy ratios, or ensure that debt covenants are met.

Question 63

An investor purchased shares on margin at €50 per share using a leverage ratio of 2.0. If the shares are sold for €70 per share, the return on the investor's equity investment is:

Correct answer: B

A. 40%. - Incorrect

Incorrect Answer Feedback: Incorrect because this ignores the leverage. The share price increased from €50 to €70 per share, (70 – 50)/50 = 40%. The leverage ratio is the ratio of the value of the position to the value of the equity investment in it.

Incorrect Answer Feedback: Incorrect because this ignores the leverage. The share price increased from €50 to €70 per share, (70 – 50)/50 = 40%. The leverage ratio is the ratio of the value of the position to the value of the equity investment in it.

Figure 57: Source equation for question 63, option A

B. 80%. - Correct

Correct Answer Feedback: Correct because the share price increased from €50 to €70 per share, (70 – 50)/50 = 40%. The leverage ratio of 2.0 results in a return on the equity investment of 2.0 × 40% = 80%. The leverage ratio is the ratio of the value of the position to the value of the equity investment in it.

Correct Answer Feedback: Correct because the share price increased from €50 to €70 per share, (70 – 50)/50 = 40%. The leverage ratio of 2.0 results in a return on the equity investment of 2.0 × 40% = 80%. The leverage ratio is the ratio of the value of the position to the value of the equity investment in it.

Figure 58: Source equation for question 63, option B

C. 180%. - Incorrect

Incorrect Answer Feedback: Incorrect because this is the return if the sale price was multiplied by the leverage ratio of 2.0, instead of multiplying the % price gain by the leverage ratio. The share price increased from €50 to €70 per share, (70 – 50)/50 = 40%. Calculating €70 × 2.0 = €140, thus (140 – 50)/50 = (90/50) = 180%. The leverage ratio is the ratio of the value of the position to the value of the equity investment in it.

Incorrect Answer Feedback: Incorrect because this is the return if the sale price was multiplied by the leverage ratio of 2.0, instead of multiplying the % price gain by the leverage ratio. The share price increased from €50 to €70 per share, (70 – 50)/50 = 40%. Calculating €70 × 2.0 = €140, thus (140 – 50)/50 = (90/50) = 180%. The leverage ratio is the ratio of the value of the position to the value of the equity investment in it.

ıo answer)

Figure 59: Source equation for question 63, option C

Question 64

With respect to asset-backed securities, which of the following most likely redistributes prepayment risk?

Correct answer: B

A. Subordination - Incorrect

Incorrect Answer Feedback: Incorrect because subordination is a form of internal credit enhancement common to securitizations. Subordination is also referred to as credit tranching. In such a structure, there is more than one bond class or tranche, and the bond classes differ as to how they will share any losses resulting from defaults of the borrowers whose loans are in the collateral. The bond classes are classified as senior bond classes or subordinated bond classes—hence, the reason this structure is also referred to as a senior/subordinated structure.

B. Time tranching - Correct

Correct Answer Feedback: Correct because time tranching is not a credit enhancement, but a method of redistributing prepayment risk, which is different than credit risk. The purpose of time tranching is the redistribution of payments based on the timing of prepayments, regardless of the credit profile. The structure of a securitization may also allow the redistribution of another type of risk, called ‘prepayment risk’ among bond classes. Prepayment risk is the uncertainty that the cash flows will be different from the scheduled cash flows as set forth in

the loan agreement because of the borrowers' ability to alter payments, usually to take advantage of interest rate movements. The creation of bond classes that possess different expected maturities is referred to as time tranching.

C. Credit tranching - Incorrect

Incorrect Answer Feedback: Incorrect because subordination is a form of internal credit enhancement common to securitizations. subordination is also referred to as credit tranching. In such a structure, there is more than one bond class or tranche, and the bond classes differ as to how they will share any losses resulting from defaults of the borrowers whose loans are in the collateral. The bond classes are classified as senior bond classes or subordinated bond classes—hence, the reason this structure is also referred to as a senior/subordinated structure.

Question 65

An investor whose portfolio lies to the right of the market portfolio on the CML most likely:

Correct answer: B

A. invested all available funds in the risk-free asset. - Incorrect

Incorrect Answer Feedback: Incorrect because this portfolio will lie to the left of the market portfolio and on the intercept of the CML.

B. borrowed funds at the risk-free rate and invested all available funds in the market portfolio. - Correct

Correct Answer Feedback: Correct because a portfolio lying to the right of the market portfolio on the CML is formed by borrowing funds at the risk-free rate and investing all available funds in the market portfolio.

C. loaned some funds at the risk-free rate and invested the remaining funds in the market portfolio. - Incorrect

Incorrect Answer Feedback: Incorrect because this portfolio will lie to the left (not right) of the market portfolio and in between the risk-free asset and the market portfolio.

Question 66

The security market line plots the expected return of a portfolio against a measure of the portfolio's:

Correct answer: B

A. total risk. - Incorrect

Incorrect Answer Feedback: Incorrect because the security market line (SML) is a graphical representation of the capital asset pricing model with beta, reflecting systematic risk (not total risk), on the x-axis and expected return on the y-axis. The security market line applies to any security, efficient or not. Total risk and systematic risk are equal only for efficient portfolios because those portfolios have no diversifiable risk remaining.

B. systematic risk. - Correct

Correct Answer Feedback: Correct because the security market line (SML) is a graphical representation of the capital asset pricing model with beta, reflecting systematic risk, on the x-axis and expected return on the y-axis.

C. unsystematic risk. - Incorrect

Incorrect Answer Feedback: Incorrect because the security market line (SML) is a graphical representation of the capital asset pricing model with beta, reflecting systematic risk (not unsystematic risk), on the x-axis and expected return on the y-axis.

Question 67

Which of the following statements is most accurate? A standard interest rate swap has:

Correct answer: A

A. a symmetric payoff profile. - Correct

Correct Answer Feedback: Correct because other similarities between interest rate forwards and swaps include the symmetric payoff profile and the fact that no cash flow is exchanged upfront. The symmetric payoff profile of a swap means that the swap has zero value to both parties at contract inception.

B. the principal cash flow exchanged upfront. - Incorrect

Incorrect Answer Feedback: Incorrect because other similarities between interest rate forwards and swaps include the symmetric payoff profile and the fact that no cash flow is exchanged upfront.

C. periodic settlements that occur at the beginning of each period. - Incorrect

Incorrect Answer Feedback: Incorrect because an FRA (forward rate agreement) has a single settlement, which occurs at the beginning of an interest period, while a standard swap has periodic settlements, which occur at the end of each respective period. So, while a standard swap does have periodic settlements, these occur at the end of the period, not the beginning.

Question 68

Which of the following is used by clearinghouses to determine the gain and loss to the parties of a futures contract?

Correct answer: C

A. Limit up and limit down - Incorrect

Incorrect Answer Feedback: Incorrect because some futures contracts contain a provision limiting price changes. These rules, called price limits, establish a band relative to the previous day's settlement price, within which all trades must occur. If market participants wish to trade at a price above the upper band, trading stops, which is called limit up, until two parties agree on a trade at a price lower than the upper limit. Likewise, if market participants wish to trade at a price below the lower band, which is called limit down, no trade can take place until two parties agree to trade at a price above the lower limit. Therefore, limit up and limit down refer to price limits but not to the practice that determines current gain and loss positions of futures contract parties.

B. Margin call - Incorrect

Incorrect Answer Feedback: Incorrect because if the futures price decreases, the long loses money and is charged with that loss, and the money is transferred to the account of the short. The account is specifically referred to as a margin account. A margin call is the practice for the contract party in a loss position to deposit more funds, after the loss position has been established through the daily settlement or mark to market practice.

C. Daily settlement - Correct

Correct Answer Feedback: Correct because identical contracts trade on an ongoing basis at different prices, reflecting the passage of time and the arrival of new information to the market. Thus, as the futures price changes, the parties make and lose money. Rising (falling) prices, of course, benefit (hurt) the long and hurt (benefit) the short. At the end of each day, the clearinghouse engages in a practice called mark to market, also known as the daily settlement.

Question 69

Which of the following statements about robo-advisers is most accurate?

Correct answer: C

A. Robo-advisers charge higher fees than traditional investment advisers - Incorrect

Incorrect Answer Feedback: Incorrect because the cost of digital investment advice provided by robo-advisers is often a fraction of traditional investment advice channels because of scalability.

B. Robo-advisers mainly target investors with a high level of investable assets - Incorrect

Incorrect Answer Feedback: Incorrect because traditional investment advice has often underserved younger and 'mass affluent' investors with lower relative levels of investable assets. Given the efficiencies of robo-advisers and the scalability of technology, customized but standardized investment advice now can be offered to a wide range and size of investors.

C. Assets managed by robo-advisers are growing in part due to new entrants such as insurance companies - Correct

Correct Answer Feedback: Correct because, reflecting low barriers to entry, large wealth management firms have introduced robo-adviser solutions to service certain customer segments and appeal to a new generation of investors. In addition to these large wealth managers, other less-traditional entrants, such as insurance companies and asset managers, are developing solutions to cross-sell into their existing clients.

Question 70

Enterprise value equals the market value of a company's debt and equity reduced by the company's:

Correct answer: B

A. debt. - Incorrect

Incorrect Answer Feedback: Incorrect because enterprise value is most frequently determined as market capitalization of equity plus market value of preferred stock plus market value of debt minus cash and investments (cash equivalents and short-term investments).

B. cash. - Correct

Correct Answer Feedback: Correct because enterprise value is most frequently determined as market capitalization of equity plus market value of preferred stock plus market value of debt minus cash and investments (cash equivalents and short-term investments).

C. preferred stock. - Incorrect

Incorrect Answer Feedback: Incorrect because enterprise value is most frequently determined as market capitalization of equity plus market value of preferred stock plus market value of debt minus cash and investments (cash equivalents and short-term investments).

Question 71

Which of the following statements is most accurate about a strong-form efficient market?

Correct answer: C

A. Fundamental analysis can generate abnormal returns - Incorrect

Incorrect Answer Feedback: Incorrect because fundamental analysis is the examination of publicly available information and the formulation of forecasts to estimate the intrinsic value of assets. However, in a strong-form efficient market, security prices fully reflect both public and private information. Therefore, fundamental analysis cannot generate abnormal return in a strong-form efficient market.

B. Insiders can earn abnormal returns from trading on private information - Incorrect

Incorrect Answer Feedback: Incorrect because in the case of a strong-form efficient market, insiders would not be able to earn abnormal returns from trading on the basis of private information. A strong-form efficient market also means that prices reflect all private information, which means that prices reflect everything that the management of a company knows about the financial condition of the company that has not been publicly released.

C. Passive portfolio management should outperform active portfolio management - Correct

Correct Answer Feedback: Correct because if securities markets are weak-form and semi-strong-form efficient, the implication is that active trading, whether attempting to exploit price patterns or public information, is not likely to generate abnormal returns. In other words, passive portfolio management should outperform active portfolio management. If this applies to weak-form and semi-strong form efficient markets, it applies to a strong-form efficient market as well.

Question 72

Commercial paper with a face value of $25,000,000 and a term of 120 days was issued 55 days ago. If the current market value is $24,855,000, the implied discount rate assuming a 360-day year is closest to:

Correct answer: A

A. 3.21%. - Correct

Correct Answer Feedback: Correct because Discount rate (DR) = (Year/Days) x ((FV PV)/FV) DR = (360/65) x ((25,000,000 24,855,000)/25,000,000) = 0.032123 3.21%.

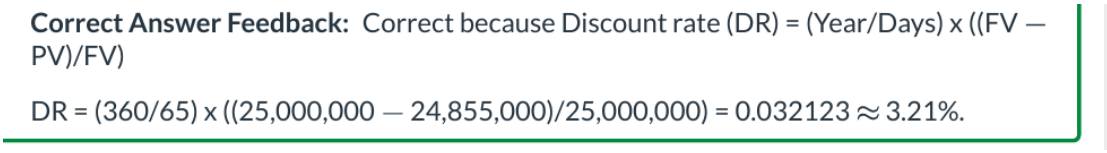


Figure 60: Source equation for question 72, option A

B. 3.23%. - Incorrect

Incorrect Answer Feedback: Incorrect because (FV PV) is divided by PV, not FV. Discount rate (DR) = (Year/Days) x ((FV PV)/FV). DR ≠ (360/65) x ((25,000,000 24,855,000)/24,855,000) = 0.03231 3.23%.

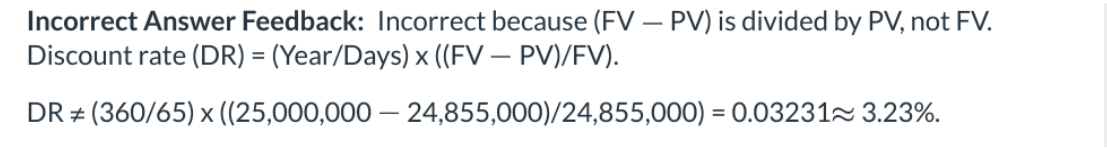


Figure 61: Source equation for question 72, option B

C. 3.80%. - Incorrect

Incorrect Answer Feedback: Incorrect because it uses the 55 days since the commercial paper was issued instead of the 65 days remaining. Discount rate (DR) = (Year/Days) x ((FV PV)/FV). DR ≠ (360/55) x ((25,000,000 24,855,000)/25,000,000) = 0.037964 3.80%.

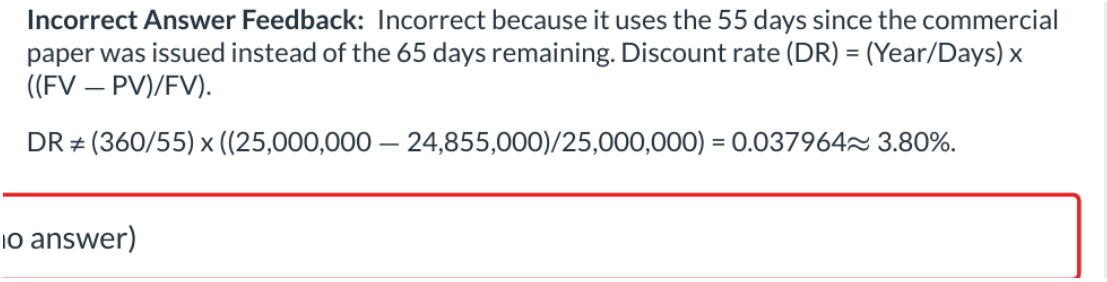


Figure 62: Source equation for question 72, option C

Question 73

All else being equal, a bond issuer is most likely to be evaluated as more creditworthy than its peers if it has a:

Correct answer: B

A. lower debt/EBITDA and a lower EBITDA/interest expense. - Incorrect

Incorrect Answer Feedback: Incorrect because lower EBITDA/interest expense indicates that the company might not be able to cover its interest payments. interest payments and thus has a higher probability of default and lower creditworthiness (all else being equal).

B. lower debt/EBITDA and a higher EBITDA/interest expense. - Correct

Correct Answer Feedback: Correct because lower debt/EBITDA shows a lower level of debt compared to adjusted earnings and due to higher EBITDA interest coverage there is less risk that the company will not be able to pay its interest (coupon), leading to a lower probability of default and higher evaluation of creditworthiness (all else being equal). Debt/EBITDA is a common leverage measure. A higher ratio indicates more leverage and thus higher credit risk. Note that this ratio can be very volatile for companies with high cash flow variability, such as those in cyclical industries and with high operating leverage (fixed costs). Coverage ratios measure an issuer's ability to meet—to 'cover'—its interest payments. The two most common are the EBITDA/interest expense and EBIT/ interest expense ratios. The ratio EBITDA/interest expense) is a bit more liberal than the ratio EBIT/interest expense because it does not subtract out the impact of (non-cash) depreciation and amortization expense. A higher ratio indicates higher credit quality.

C. higher debt/EBITDA and a lower EBITDA/interest expense. - Incorrect

Incorrect Answer Feedback: Incorrect because higher debt/EBITDA shows that there is higher leverage in comparison to earnings and this increases the probability of default and lowers the evaluation of creditworthiness of the company (all else being equal) whilst a lower EBITDA/interest expense indicates that the company might not be able to cover its interest payments. interest payments and thus has a higher probability of default and lower creditworthiness (all else being equal).

Question 74

An analyst gathers the following information about a hedge fund's fee structure:

Assets under management (AUM), beginning of period	$80 million
Management fee, based on year-end AUM	2%
Incentive fee	20%

The management and incentive fees are calculated independently. If the hedge fund generates a gross return of 25%, the assets under management after fees at the end of the period are closest to:

Correct answer: A

A. $94.0 million. - Correct

Correct Answer Feedback: Correct because management fee = $80 million × (1 + 25%) × 2% = $2 million; Incentive fee = ($80 million × 25%) × 20% = $4 million; Assets under management (AUM), net of fees = ($80 million × (1 + 25%) − $2 million − $4 million) = $94.0 million.

Correct Answer Feedback: Correct because management fee = $80 million × (1 + 25%) × 2% = $2 million;
Incentive fee = ($80 million × 25%) × 20% = $4 million;
Assets under management (AUM), net of fees = ($80 million × (1 + 25%) − $2 million − $4 million) = $94.0 million.

Figure 63: Source equation for question 74, option A

B. $94.4 million. - Incorrect

Incorrect Answer Feedback: Incorrect because incentive fees are considered to be calculated net of management fees. Management fee = $80 million × (1 + 25%) × 2% = $2 million; Incentive fee ≠ (($80 million × 25%) − $2 million) × 20% = $3.6 million; Assets under management (AUM), net of fees ≠ ($80 million × (1 + 25%) − $2 million − $3.6 million) = $94.4 million.

Incorrect Answer Feedback: Incorrect because incentive fees are considered to be calculated net of management fees.

Figure 64: Source equation for question 74, option B

Management fee = $80 million × (1 + 25%) × 2% = $2 million;
Incentive fee ≠ (($80 million × 25%) − $2 million) × 20% = $3.6 million;
Assets under management (AUM), net of fees ≠ ($80 million × (1 + 25%) − $2 million − $3.6 million) = $94.4 million.

Figure 65: Source equation for question 74, option B

C. $97.6 million. - Incorrect

Incorrect Answer Feedback: Incorrect because it is the result of multiplying the total fees plus 1 with the initial investment: Assets under management (AUM), net of fees ≠ $80 million × (1 + 20% + 2%) = $97.6 million.

Incorrect Answer Feedback: Incorrect because it is the result of multiplying the total fees plus 1 with the initial investment: Assets under management (AUM), net of fees ≠ $80 million × (1 + 20% + 2%) = $97.6 million.

ıo answer)

Figure 66: Source equation for question 74, option C

Question 75

An analyst gathers the following information about two companies' non-callable, non- convertible perpetual preference shares:

	Preference Share 1	Preference Share 2
Par value	$100	$100
Required rate of return	8%	11%
Annual dividend per share	$6	$8

The value of Preference Share 1 is:

Correct answer: C

A. less than the value of Preference Share 2. - Incorrect

Incorrect Answer Feedback: Incorrect because the present value (PV) of Preference Share 1 is greater than the PV of Preference Share 2. That is, the present value (PV) of a non-callable, non-convertible, perpetual preference share is solved using the equation: V = D / r, where: D = current dividend; r = required rate of return on the 0 0 0 stock. Therefore the PV of Preference Share 1 = $6 / 0.08% = $75.00, which is greater than the PV of Preference Share 2 = $8 / 0.11 ≈ $72.73.

B. equal to the value of Preference Share 2. - Incorrect

Incorrect Answer Feedback: Incorrect because the present value (PV) of Preference Share 1 is greater than the PV of Preference Share 2. That is, the present value (PV) of a non-callable, non-convertible, perpetual preference share is solved using the equation: V = D / r, where: D = current dividend; r = required rate of return on the 0 0 0 stock. Therefore the PV of Preference Share 1 = $6 / 0.08 = $75.00, which is greater than the PV of Preference Share 2 = $8 / 0.11 ≈ $72.73.

C. greater than the value of Preference Share 2. - Correct

Correct Answer Feedback: Correct because the present value (PV) of a non-callable, non-convertible, perpetual preference share is solved using the equation: V = D /r, 0 0 (where: D = current dividend; r = required rate of return on the stock. Therefore the 0 PV of Preference Share 1 = $6 / 0.08 = $75.00, which is greater than the PV of Preference Share 2 = $8 / 0.11 ≈ $72.73.

Incorrect Answer Feedback: Incorrect because the present value (PV) of Preference Share 1 is greater than the PV of Preference Share 2. That is, the present value (PV) of a non-callable, non-convertible, perpetual preference share is solved using the equation: $V_0 = D_0 / r$, where: D_0 = current dividend; r = required rate of return on the stock. Therefore the PV of Preference Share 1 = \$6 / 0.08% = \$75.00, which is greater than the PV of Preference Share 2 = \$8 / 0.11 ≈ \$72.73.

Figure 67: Source equation for question 75, option A

Incorrect Answer Feedback: Incorrect because the present value (PV) of Preference Share 1 is greater than the PV of Preference Share 2. That is, the present value (PV) of a non-callable, non-convertible, perpetual preference share is solved using the equation: $V_0 = D_0 / r$, where: D_0 = current dividend; r = required rate of return on the stock. Therefore the PV of Preference Share 1 = \$6 / 0.08 = \$75.00, which is greater than the PV of Preference Share 2 = \$8 / 0.11 ≈ \$72.73.

Figure 68: Source equation for question 75, option B

Correct Answer Feedback: Correct because the present value (PV) of a non-callable, non-convertible, perpetual preference share is solved using the equation: $V_0 = D_0 / r$.

Figure 69: Source equation for question 75, option C

(where: D_0 = current dividend; r = required rate of return on the stock. Therefore the PV of Preference Share 1 = \$6 / 0.08 = \$75.00, which is greater than the PV of Preference Share 2 = \$8 / 0.11 ≈ \$72.73.

ıo answer)

Figure 70: Source equation for question 75, option C

Based only on this information, the estimated market discount rate for a 6-year bond with similar credit quality is:

Correct answer: C

A. 3.9%. - Incorrect

Incorrect Answer Feedback: Incorrect this response is an incorrect application of matrix pricing, where the estimated market discount rate can be obtained with linear interpolation. This response is incorrect because it uses linear interpolation from the wrong end between the two given bonds: 0.033 + (7 – 6) / (7 – 4) × (0.051 – 0.033) = 0.033 + 1 / 3 × 0.018 = 0.039 = 3.9%.

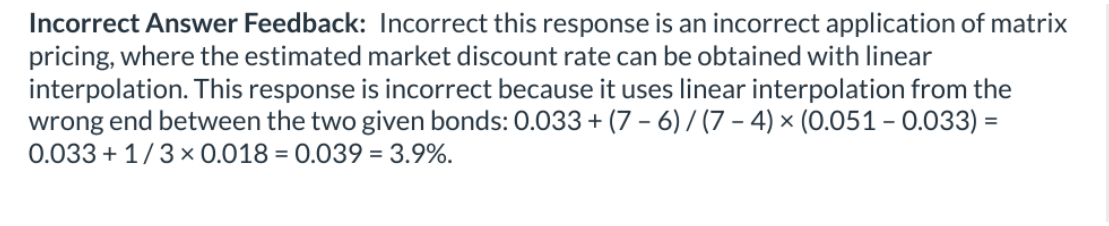


Figure 71: Source equation for question 76, option A

B. 4.2%. - Incorrect

Incorrect Answer Feedback: Incorrect this response is an incorrect application of matrix pricing, where the estimated market discount rate can be obtained with linear interpolation. This response is incorrect because it is the simple average of the two given bonds: (0.033 + 0.051) / 2 = 0.084 / 2 = 0.042 = 4.2%.

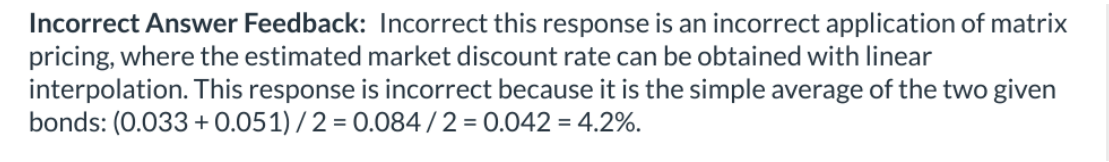


Figure 72: Source equation for question 76, option B

C. 4.5%. - Correct

Correct Answer Feedback: Correct because some fixed-rate bonds are not actively traded. Therefore, there is no market price available to calculate the rate of return required by investors. In these situations, it is common to estimate the market discount rate and price based on the quoted or flat prices of more frequently traded comparable bonds. These comparable bonds have similar times-to-maturity, coupon rates, and credit quality. This estimation process is called matrix pricing. The estimated market discount rate can be obtained with linear interpolation. Using linear interpolation between the two given bonds, we have: 0.033 + (6 – 4) / (7 – 4) × (0.051 – 0.033) = 0.033 + 2 / 3 × 0.018 = 0.045 = 4.5%.

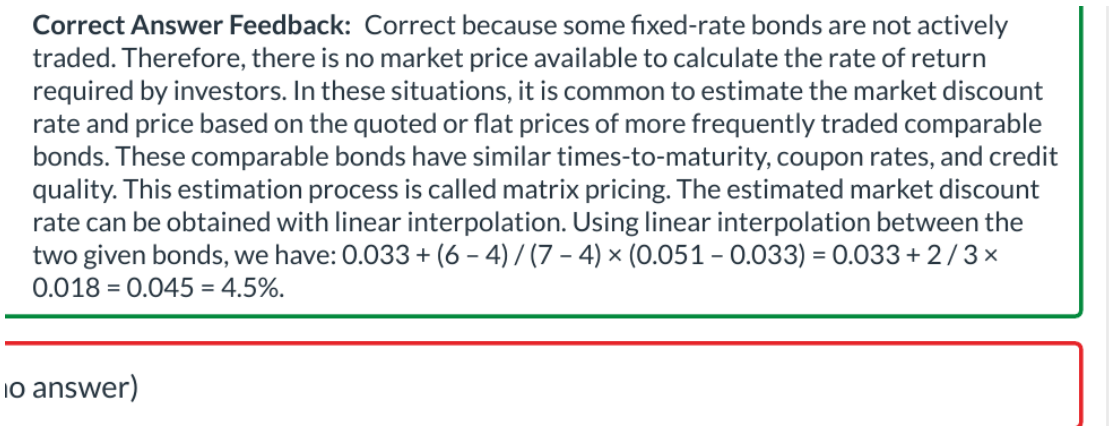


Figure 73: Source equation for question 76, option C

Question 77

Which of the following statements best describes changes in the value of a long forward position during its life?

Correct answer: C

A. As interest rates go down, the value of the position goes up - Incorrect

Incorrect Answer Feedback: Incorrect because as interest rates go down, the value of the contract goes down.

B. As the time to maturity goes down, the value of the position goes up - Incorrect

Incorrect Answer Feedback: Incorrect because as the time to maturity goes down, the value of the contract goes down.

C. As the price of the underlying goes up, the value of the position goes up - Correct

Correct Answer Feedback: Correct because given the formula for the value of a forward contract: –(T–t) V (T) = S – F (T)(1 + r) it follows that the value of the contract goes t t 0 up as the price of the underlying goes up.

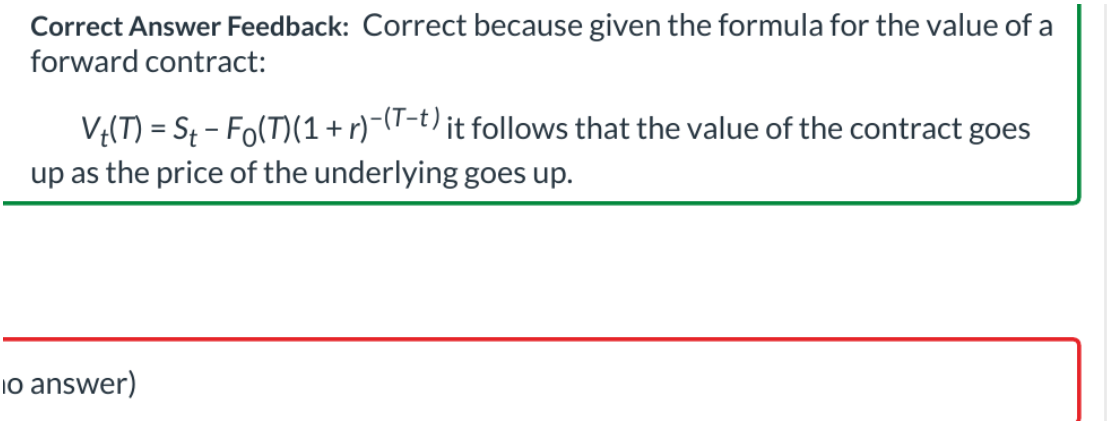


Figure 74: Source equation for question 77, option C

Question 78

A reason for a company's management to repurchase its shares is to signal a belief that the company's shares are:

Correct answer: A

A. undervalued. - Correct

Correct Answer Feedback: Correct because company managements have expressed several key reasons for engaging in share repurchases— namely, signaling a belief that their shares are undervalued.

B. fairly valued. - Incorrect

Incorrect Answer Feedback: Incorrect because company managements have expressed several key reasons for engaging in share repurchases— namely, signaling a belief that their shares are undervalued, not fairly valued.

C. overvalued. - Incorrect

Incorrect Answer Feedback: Incorrect because company managements have expressed several key reasons for engaging in share repurchases— namely, signaling a belief that their shares are undervalued, not overvalued.

Question 79

An analyst gathers the following estimates about a company:

Earnings retention rate	40%
Growth rate	3%

If the justified forward P/E is 10 based on the Gordon growth model, the required rate of return is:

Correct answer: C

A. 6%. - Incorrect

Incorrect Answer Feedback: Incorrect because it is the result when the second term, g, is omitted: r = p / 10 + g = 0.6 / 10 + 0 = 0.06 = 6%.

Incorrect Answer Feedback: Incorrect because it is the result when the second term, *g*, is omitted: *r* = p / 10 + g = 0.6 / 10 + 0 = 0.06 = 6%.

Figure 75: Source equation for question 79, option A

B. 7%. - Incorrect

Incorrect Answer Feedback: Incorrect because it is the result when the retention rate is substituted for the payout rate: r = p / 10 + g = 0.4 / 10 + 0.03 = 0.07 = 7%.

Incorrect Answer Feedback: Incorrect because it is the result when the retention rate is substituted for the payout rate:

r = *p* / 10 + *g* = 0.4 / 10 + 0.03 = 0.07 = 7%.

Figure 76: Source equation for question 79, option B

C. 9%. - Correct

Correct Answer Feedback: Correct because 9% is the required rate of return (r), determined by rearranging the justified forward P/E ratio to solve for r, which requires first calculating the dividend payout ratio. The dividend payout ratio = (1 – earnings retention rate) = (1 – 0.40) = 0.60. The justified forward P/E = (D / E) / (r–g) = p / (r – 1 1 g) = 10. Rearranging the equation: r = p / (P/E) + g = (0.6 / 10) + 0.03 = 0.09 = 9%, where P = current intrinsic value, E = forecast for next year's earnings, p = dividend 0 1 payout ratio, g = growth rate.

Correct Answer Feedback: Correct because 9% is the required rate of return (*r*), determined by rearranging the justified forward P/E ratio to solve for *r*, which requires first calculating the dividend payout ratio. The dividend payout ratio = (1 – earnings retention rate) = (1 – 0.40) = 0.60. The justified forward P/E = (D_1 / E_1) / (*r*–*g*) = *p* / (*r* – *g*) = 10. Rearranging the equation: *r* = *p* / (P/E) + *g* = (0.6 / 10) + 0.03 = 0.09 = 9%, where P_0 = current intrinsic value, E_1 = forecast for next year's earnings, *p* = dividend payout ratio, *g* = growth rate.

Figure 77: Source equation for question 79, option C

Question 80

Which of the following is most likely considered an affirmative covenant?

Correct answer: C

A. A minimum interest coverage ratio - Incorrect

Incorrect Answer Feedback: Incorrect because this is a negative covenant. Bondholders are protected by a set of covenants which closely monitor certain financial ratios and restrict its ability to pay dividends to shareholders, repurchase shares, and / or take on additional debt. Negative covenants include limitations on investments, the disposal of assets, or issuance of debt senior to existing obligations under what is known as a negative pledge clause. Negative covenants seek to ensure that an issuer maintains the ability to make interest and principal payments.

B. A limit on the level of share buybacks - Incorrect

Incorrect Answer Feedback: Incorrect because this is a negative covenant. Bondholders are protected by a set of covenants which closely monitor certain financial ratios and restrict its ability to pay dividends to shareholders, repurchase shares, and / or take on additional debt.

C. What the issuer will do with the proceeds of a bond issue - Correct

Correct Answer Feedback: Correct because this is an affirmative covenant. Bondholders have limited influence over an issuer as compared to equity investors, who have voting rights. One exception to this is legally enforceable rules or bond covenants that borrowers and lenders agree on at the time a bond is issued or a loan is made. Affirmative covenants specify what issuers are required to do, whereas negative covenants specify what issuers are prohibited from doing. Affirmative covenants are typically administrative in nature. For example, common affirmative covenants include the use of proceeds from the bond issue, the provision of timely financial reports, and permitting bondholders to redeem their bonds at a premium to par if the issuer is acquired.

Question 81

The actions of non-hedging investors impact:

Correct answer: C

A. commodity supply only. - Incorrect

Incorrect Answer Feedback: Incorrect because non-hedging investors can also affect demand for commodities. Demand for commodities is determined by the needs of end users and secondarily by the actions of non-hedging investors.

B. commodity demand only. - Incorrect

Incorrect Answer Feedback: Incorrect because non-hedging investors can also affect supply of commodities. Supplies of physical commodities are determined by production and inventory levels and secondarily by the actions of non-hedging investors.

C. both commodity supply and commodity demand. - Correct

Correct Answer Feedback: Correct because supplies of physical commodities are determined by production and inventory levels and secondarily by the actions of non- hedging investors. Demand for commodities is determined by the needs of end users and secondarily by the actions of non-hedging investors.

Question 82

Accounting standards and reporting requirements that produce meaningful and timely financial disclosures are most critical for achieving which of the following efficiencies associated with a well-functioning financial system?

Correct answer: B

A. Allocational - Incorrect

Incorrect Answer Feedback: Incorrect. Allocational efficiency refers to making resources available where they are most valuable.

B. Informational - Correct

Correct Answer Feedback: Correct. Accounting standards and reporting requirements that allow meaningful and timely financial disclosures reduce the costs of obtaining fundamental information and thereby allow analysts to form more accurate estimates of fundamental values. They support informationally efficient markets.

C. Operational - Incorrect

Incorrect Answer Feedback: Incorrect. Operational efficiency relates to the costs of arranging trades, which can be reduced via organized exchanges, brokerages, securitization, clearing houses, and so forth.

Question 83

At initiation, the price of a forward contract is most likely:

Correct answer: C

A. less than the value of the forward contract. - Incorrect

Incorrect Answer Feedback: Incorrect because, as described in the response rationale for the correct answer, the price of a forward contract at initiation is not less than the value of the forward contract.

B. equal to the value of the forward contract. - Incorrect

Incorrect Answer Feedback: Incorrect because, as described in the response rationale for the correct answer, the price of a forward contract at initiation is not equal to the value of the forward contract.

C. greater than the value of the forward contract. - Correct

Correct Answer Feedback: Correct because neither the long nor the short pays anything to the other at the initiation date of a forward contract, the value of a forward contract when initiated is zero. On the other hand, the forward price that the parties agree to at the initiation date of the contract is a special price that results in the contract having zero value and prohibiting arbitrage. As this price agreed by the parties is a positive number, the price of a forward contract at initiation is greater than the value of the contract, which is zero.

Question 84

An asset-backed securitization with a waterfall structure is most likely structured with which type of credit enhancement?

Correct answer: A

A. Subordination - Correct

Correct Answer Feedback: Correct. Asset-backed securitizations can be structured with subordinated bond classes. They function as credit protection for the more senior bond classes; that is, losses are realized by the subordinated bond classes before any losses are realized by the senior bond classes. This type of protection is also commonly referred to as a waterfall structure because of the cascading flow of payments between bond classes in the event of default.

B. Time tranching - Incorrect

Incorrect Answer Feedback: Incorrect. The creation of bond classes that possess different expected maturities is referred to as time tranching.

C. Special-purpose entity (SPE) - Incorrect

Incorrect Answer Feedback: Incorrect. An SPE is a bankruptcy remote legal entity that holds the collateral and is considered not an enhancement but, rather, a prerequisite for establishing securitizations.

Question 85

Failed market regulation most likely results in decreased:

Correct answer: B

A. fraud. - Incorrect

Incorrect Answer Feedback: Incorrect because financial markets tend to be complex, and because customers are often much less sophisticated than the professionals that serve them, the potential for losses through various frauds can be unacceptably high in unregulated markets.

B. economic efficiency. - Correct

Correct Answer Feedback: Correct because economies that cannot solve regulatory problems tend to operate less efficiently than do better regulated economies, and they tend to be less wealthy.

C. costs for investor participation. - Incorrect

Incorrect Answer Feedback: Incorrect because regulation is necessary because regulating certain behaviors through market-based mechanisms is too costly for people who are unsophisticated and uninformed.

Question 86

Under which of the following alternative investment methods can an investor invest indirectly in alternative assets?

Correct answer: C

A. Co-investing only - Incorrect

Incorrect Answer Feedback: Incorrect because under co-investing, the investor invests in assets indirectly through the fund but also possesses rights (known as co-investment rights) to invest directly in the same assets.

B. Fund investing only - Incorrect

Incorrect Answer Feedback: Incorrect because fund investing can be viewed as an indirect method of investing in alternative assets. In addition, under co-investing, the investor invests in assets indirectly through the fund but also possesses rights (known as co-investment rights) to invest directly in the same assets.

C. Both co-investing and fund investing - Correct

Correct Answer Feedback: Correct because fund investing can be viewed as an indirect method of investing in alternative assets. In addition, under co-investing, the investor invests in assets indirectly through the fund but also possesses rights (known as co- investment rights) to invest directly in the same assets.

Question 87

Information for Stock A and the market appear in the following table:

Standard deviation of Stock A's return	40%
Standard deviation of the market's return	20%
Correlation of Stock A with the market	85%

The beta of Stock A is closest to:

Correct answer: B

A. 0.43. - Incorrect

Incorrect Answer Feedback: Incorrect because this calculation incorrectly swaps the standard deviation of the market and the standard deviation of the stock between the numerator and denominator: $0.85 \times 0.20 \div 0.40 = 0.43$

Incorrect Answer Feedback: Incorrect because this calculation incorrectly swaps the standard deviation of the market and the standard deviation of the stock between the numerator and denominator: 0.85 × 0.20÷0.40 = 0.43

Figure 78: Source equation for question 87, option A

B. 1.70. - Correct

Correct Answer Feedback: Correct. Beta, β, of a stock is calculated as: β = ρ σ ÷ σ i,m i m where ρ = correlation of the stock with the market i,m σ = standard deviation of the stock i σ = standard deviation of the market m Plugging in the numbers β = 0.85×0.40÷0.20 = 1.70.

Correct Answer Feedback: Correct. Beta, β, of a stock is calculated as:

$\beta = \rho_{i,m}\sigma_i \div \sigma_m$

where

- $\rho_{i,m}$ = correlation of the stock with the market
- σ_i = standard deviation of the stock
- σ_m = standard deviation of the market

Plugging in the numbers

$\beta = 0.85\times0.40\div0.20 = 1.70.$

Figure 79: Source equation for question 87, option B

C. 2.35. - Incorrect

Incorrect Answer Feedback: Incorrect because this calculation incorrectly puts the correlation in the denominator: 0.4÷(0.2×0.85) = 2.35.

Incorrect Answer Feedback: Incorrect because this calculation incorrectly puts the correlation in the denominator: 0.4÷(0.2×0.85) = 2.35.

ıo answer)

Figure 80: Source equation for question 87, option C

Question 88

Which of the following is most likely a cross-sectional anomaly in financial markets?

Correct answer: A

A. Value effect - Correct

Correct Answer Feedback: Correct because the value effect—that is, stocks with below-average price-to-earnings and market-to-book ratios and above- average dividend yields have consistently outperformed growth stocks over long periods—is a cross-sectional anomaly.

B. Overreaction effect - Incorrect

Incorrect Answer Feedback: Incorrect because the overreaction effect is a time- series anomaly, not a cross-sectional anomaly.

C. Closed-end fund discount - Incorrect

Incorrect Answer Feedback: Incorrect because a closed-end fund discount is an anomaly where a closed-end fund trades at a discount from its net asset value. It is not a cross-sectional anomaly.

Question 89

Compared to investing in a single security, the portfolio approach can improve investors' risk–return trade-off by offering:

Correct answer: B

A. lower expected return with the same overall volatility of returns. - Incorrect

Incorrect Answer Feedback: Incorrect because in addition to avoiding a potential disaster associated with over investing in a single security, portfolios also generally offer equivalent expected returns with lower overall volatility of returns—as represented by a measure such as standard deviation.

B. the same expected return with lower overall volatility of returns. - Correct

Correct Answer Feedback: Correct because in addition to avoiding a potential disaster associated with over investing in a single security, portfolios also generally offer equivalent expected returns with lower overall volatility of returns—as represented by a measure such as standard deviation. One of the critical ideas about portfolios: Portfolios affect risk more than returns.

C. higher overall volatility of returns for a given level of expected return. - Incorrect

Incorrect Answer Feedback: Incorrect because in addition to avoiding a potential disaster associated with over investing in a single security, portfolios also generally offer equivalent expected returns with lower (not higher) overall volatility of returns—as represented by a measure such as standard deviation. One of the critical ideas about portfolios: Portfolios affect risk more than returns.

Question 90

In a rising interest rate environment, the effective duration of a putable bond relative to an otherwise identical non-putable bond, will most likely be:

Correct answer: A

A. lower. - Correct

Correct Answer Feedback: Correct because when interest rates are rising, the put option becomes more valuable to the investor. The ability to sell the bond at par value limits the price depreciation as rates rise. So, the presence of an embedded put option reduces the sensitivity of the bond price to changes in interest rates, resulting in a lower effective duration.

B. the same. - Incorrect

Incorrect Answer Feedback: Incorrect because in a rising interest rate environment the effective duration of a putable bond will be lower than the effective duration of a comparable non-putable bond.

C. higher. - Incorrect

Incorrect Answer Feedback: Incorrect because in a rising interest rate environment the effective duration of a putable bond will be lower, not higher, than the effective duration of a comparable non-putable bond.